

Particle methods in KRATOS

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IS KRATOS “JUST” A FINITE ELEMENT PLATFORM?



....this was the initial idea many years ago...

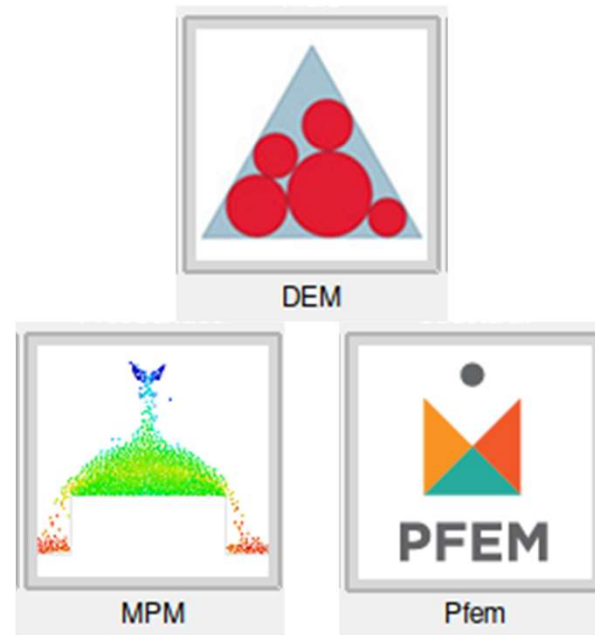
BUT
the answer is **NO**,
KRATOS IS NOT JUST FINITE ELEMENTS!



KRATOS IS ALSO A PLATFORM FOR PARTICLE METHODS

Particle methods can be divided in:

- Discrete (DEM)
- Continuum based (PFEM, MPM)



Particle methods. The DEM application

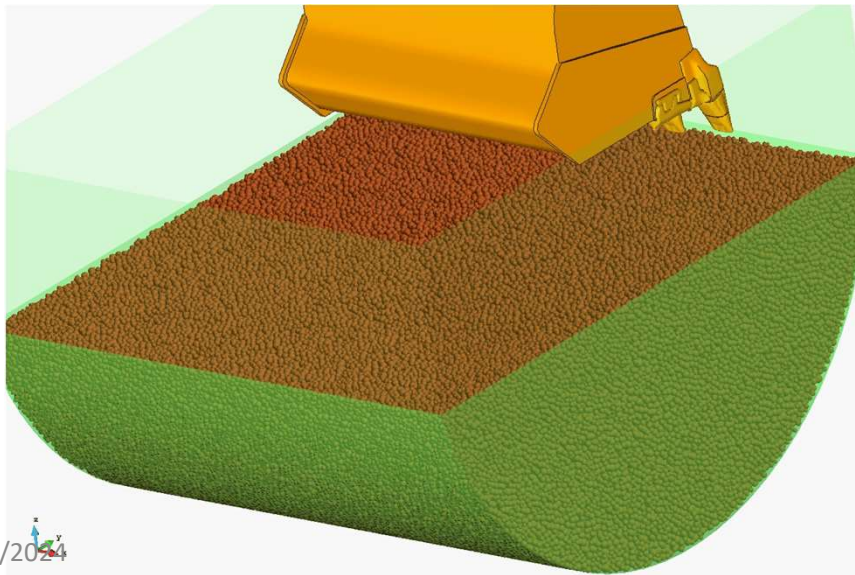
Contact: Guillermo Casas gcasas@cimne.upc.edu



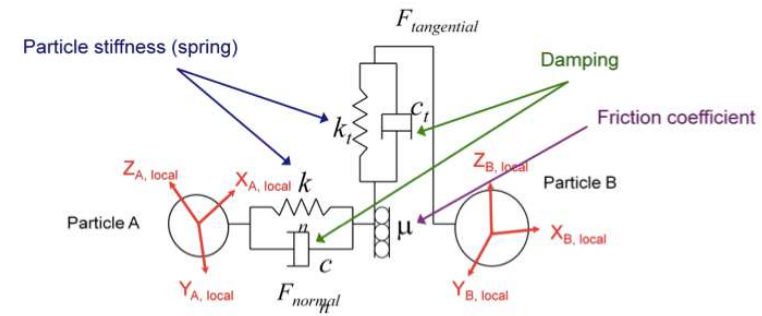
DEM

- Cohesive and non-cohesive materials
- Spherical and non-spherical particles
- Parallel computing
- Several contact laws
- Python interface
- Easy/fast coupling with other Kratos apps

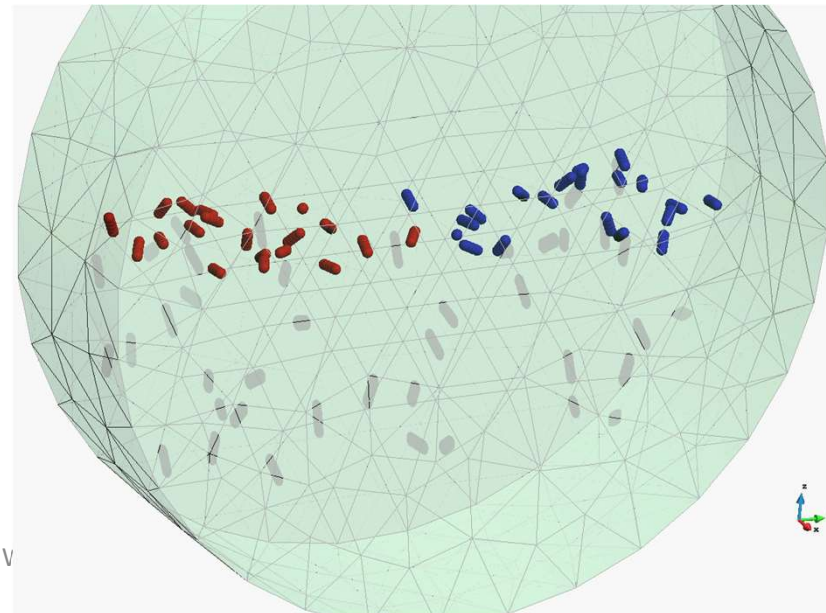
Spherical particles



6 – 8/11/2024

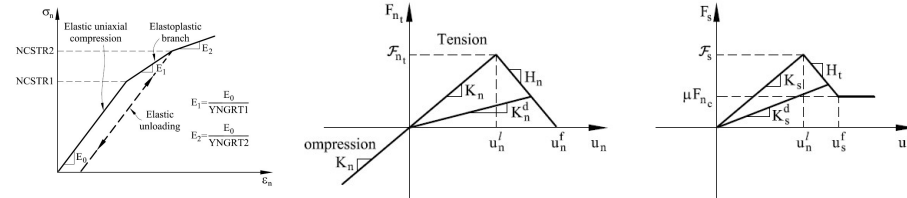


Non-spherical particles

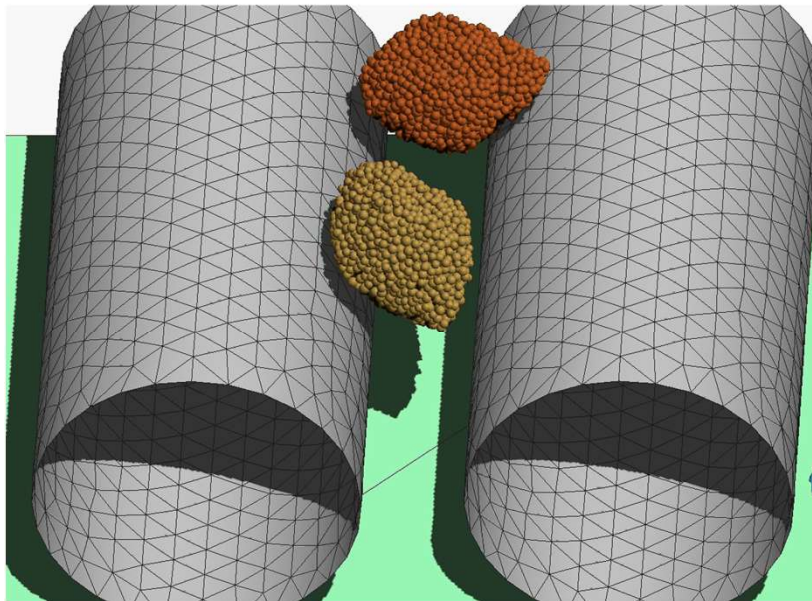


DEM

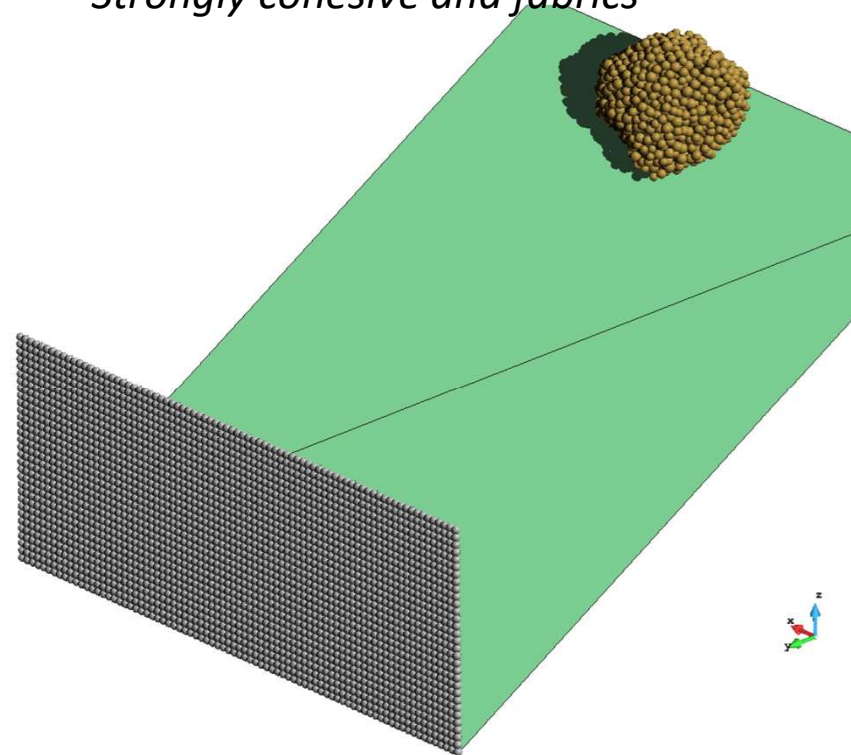
- Bonded particles
- Custom constitutive law
- Bond damage, plasticity, fracture, ...
- Sintering

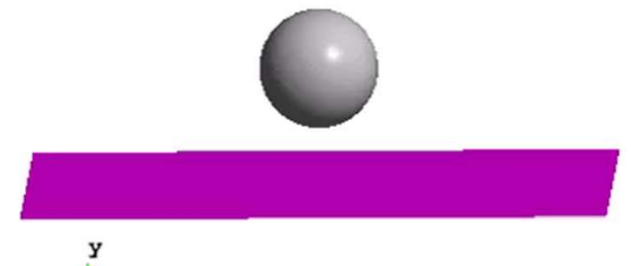
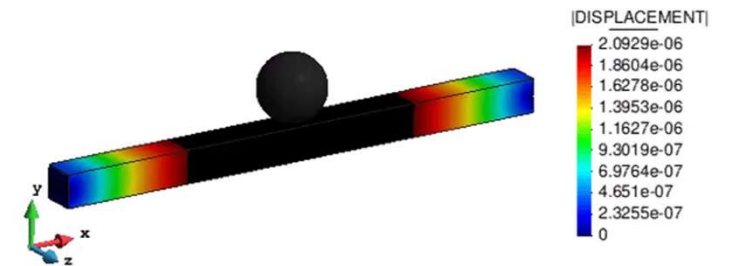
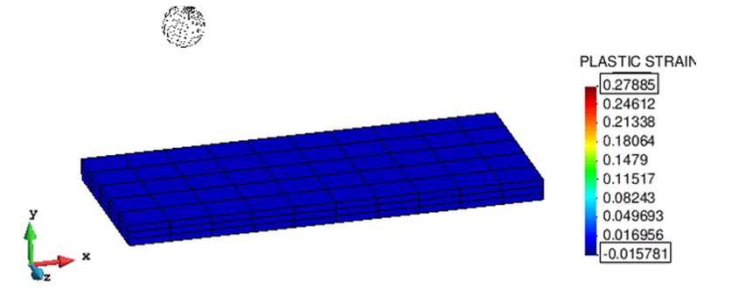
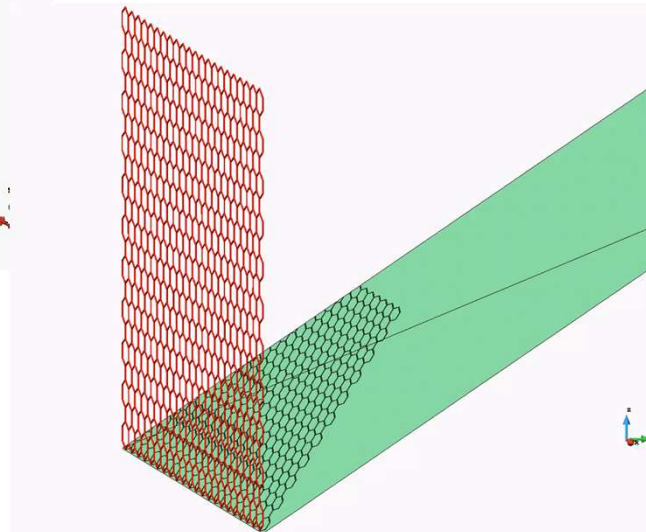
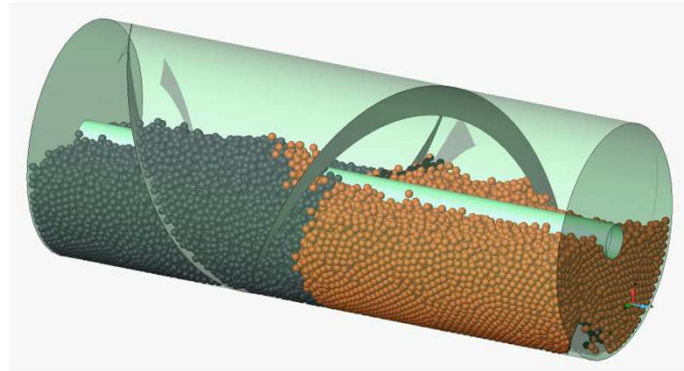
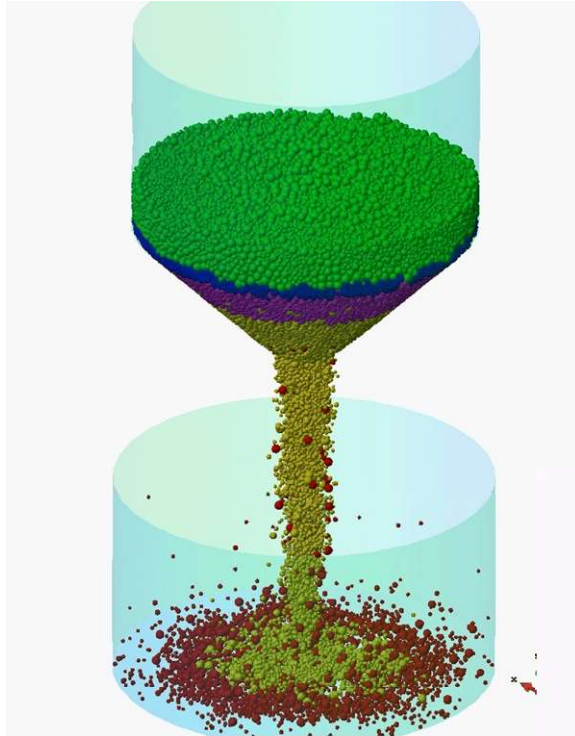


Slightly cohesive

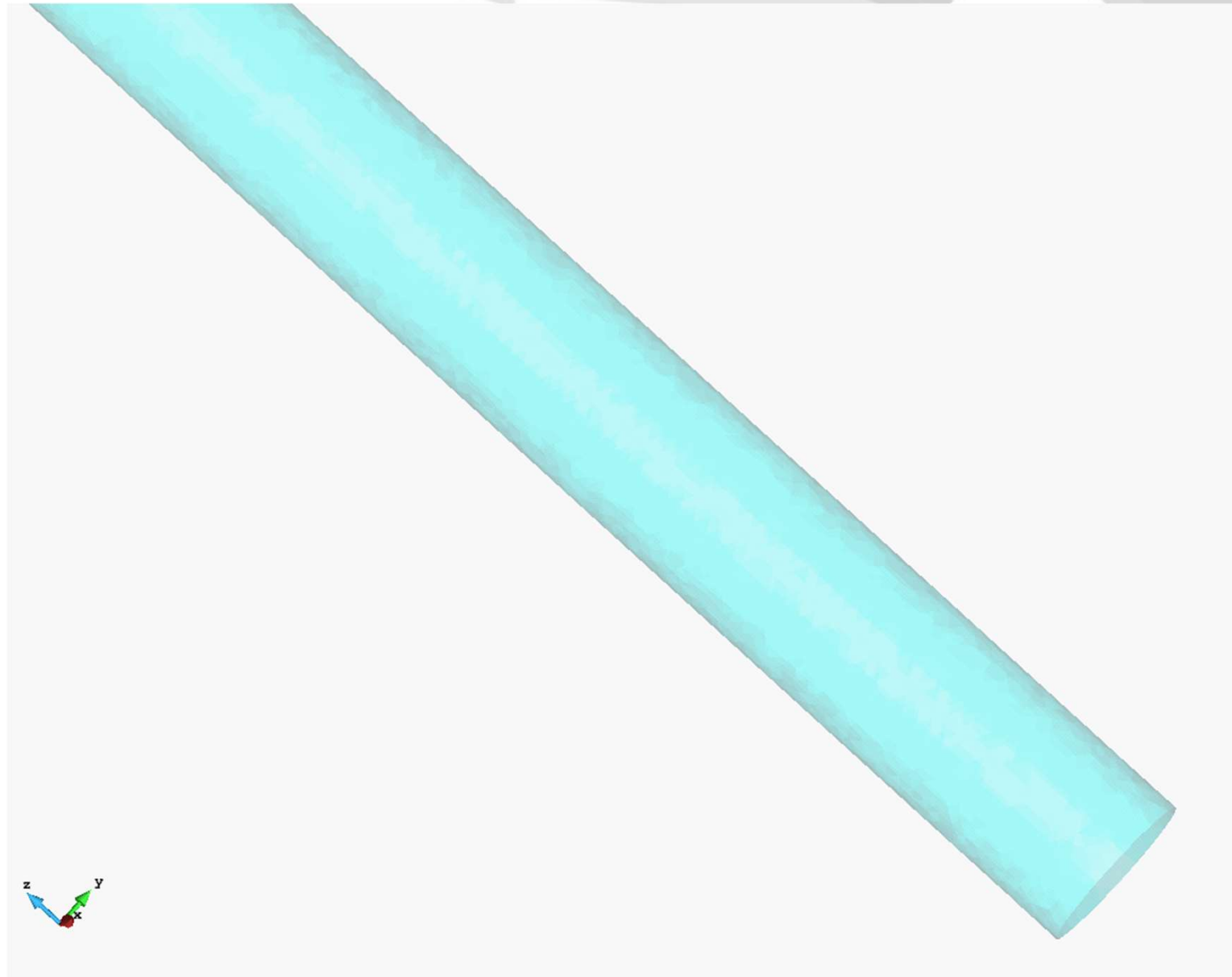


Strongly cohesive and fabrics





Coupled DEF-CFD



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Particle methods. The PFEM application

Contact: Alessandro Franci



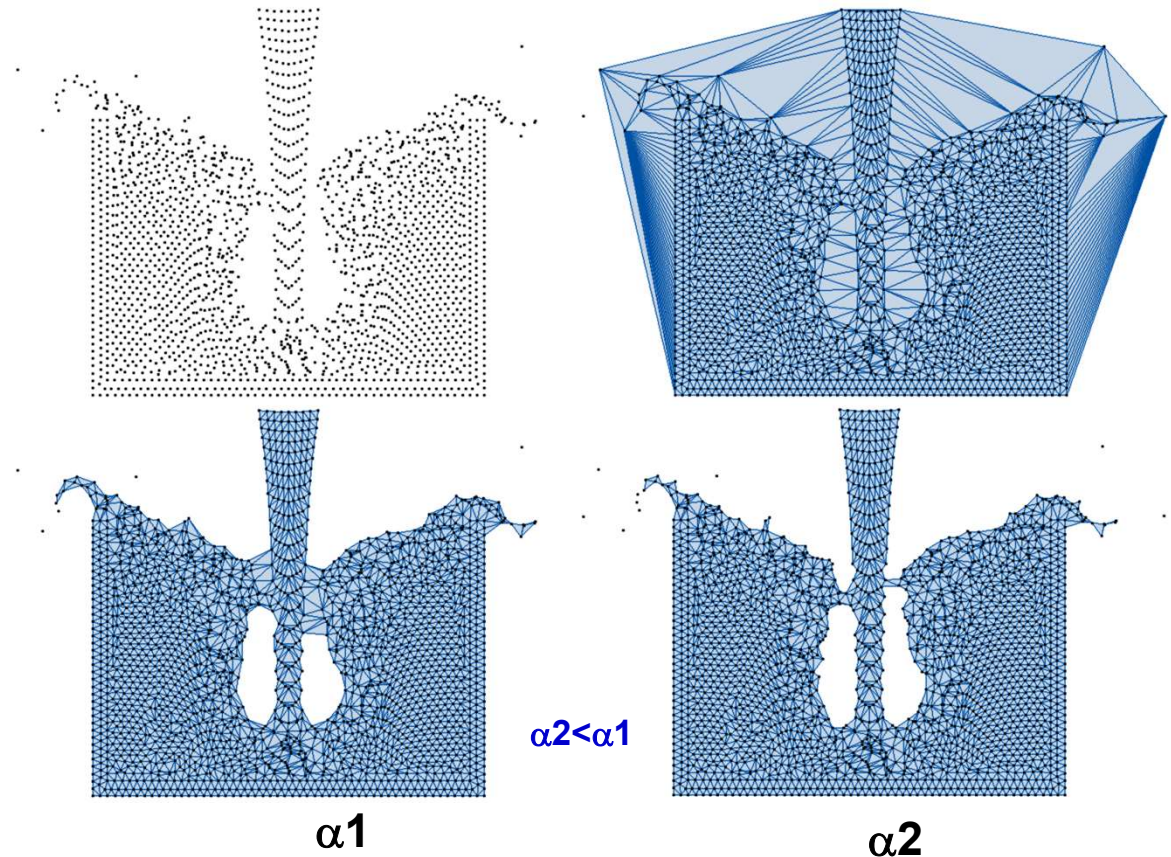
PFEM

Particle Finite Element Method PFEM

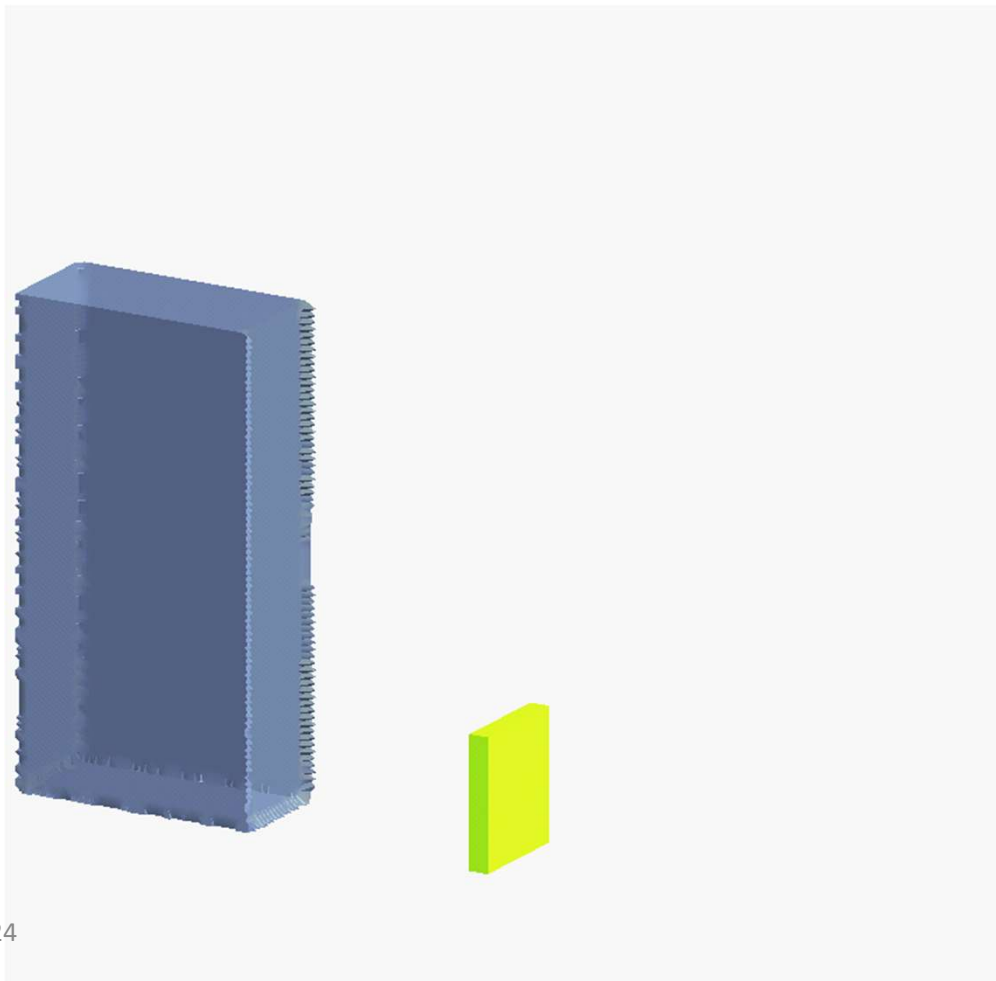
(<http://www.cimne.com/pfem>)

Firstly proposed by Prof S. Idelsohn and E. Onate for simulating breaking waves.

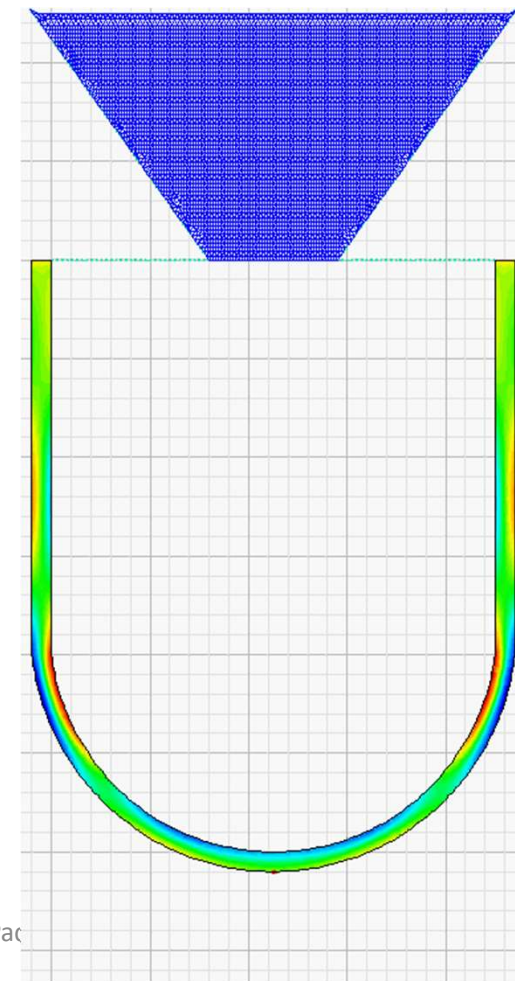
- Finite Element Method (**FEM**);
- **Updated LAGRANGIAN** kinematical description of motion
- **REMESHING** algorithm
- **ALPHA SHAPE** to detect the free surface



PFEM for FSI



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- Pac



Particle methods. The MPM application

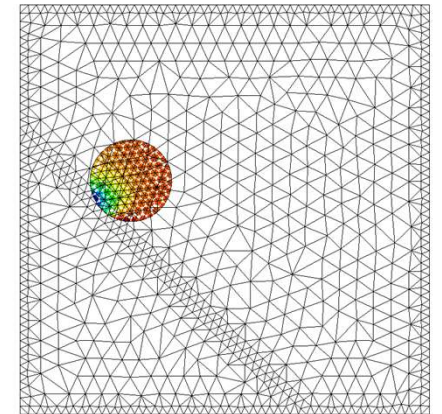
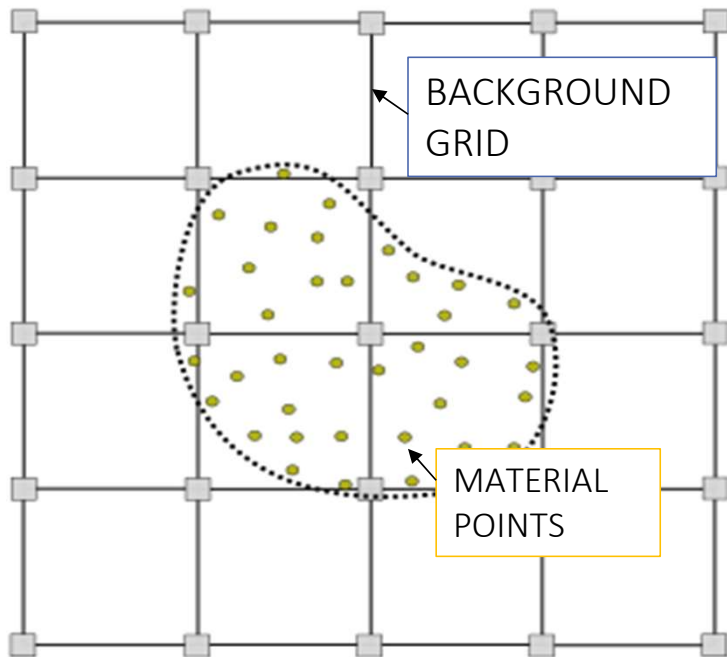


The MATERIAL POINT METHOD as an “enriched” FEM

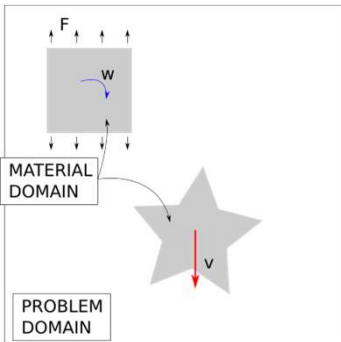
Numerical technique proposed for the first time by [Harlow, 1964]

The Material Point Method uses two different discretizations:

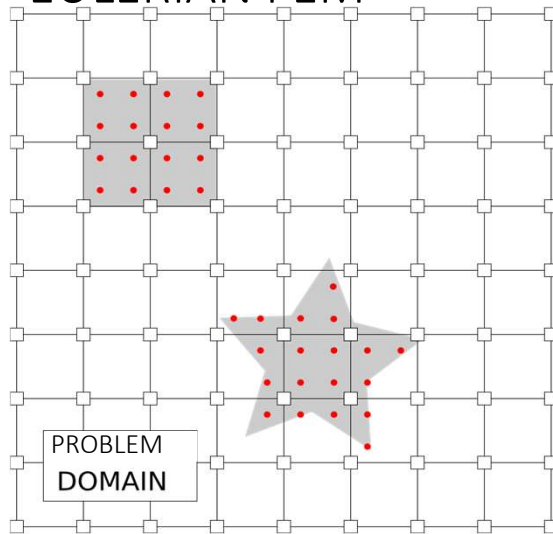
- A **BACKGROUND GRID** (fixed)
 - Used for the FEM solution system
- A collection of **MATERIAL POINTS (MP)** (Lagrangian)
 - Moving integration points
 - MPs store historical information



FEM

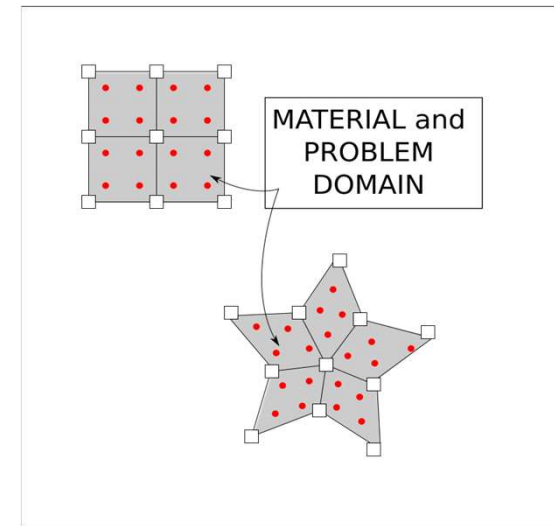


EULERIAN FEM



- ✓ The nodes are SPATIAL points
 - Advection is present
 - Tracking techniques needed for interfaces (level set, ...)
 - Complex imposition of BCs on interfaces
- ✓ No mesh distortion (fixed grid)
 - More diffusive

LAGRANGIAN FEM

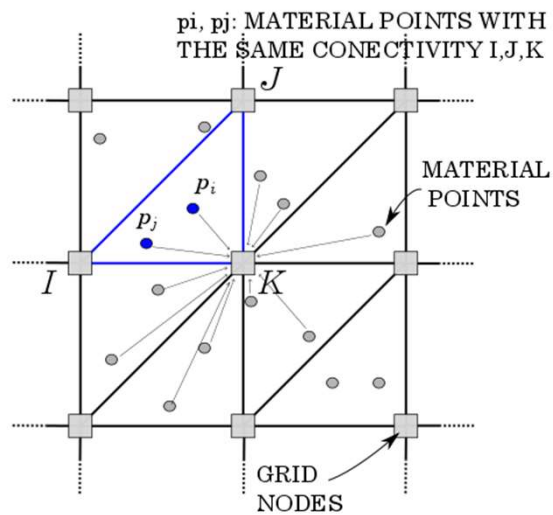


- ✓ The nodes are MATERIAL points
 - ✓ No advection
 - ✓ Natural treatment of material interfaces
 - ✓ Easy imposition of BCs
- Mesh distortion
- Volume conservation issues



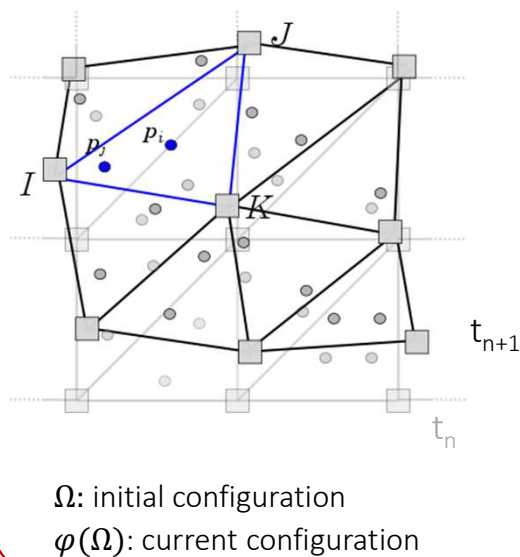
MPM steps

INITIALIZATION



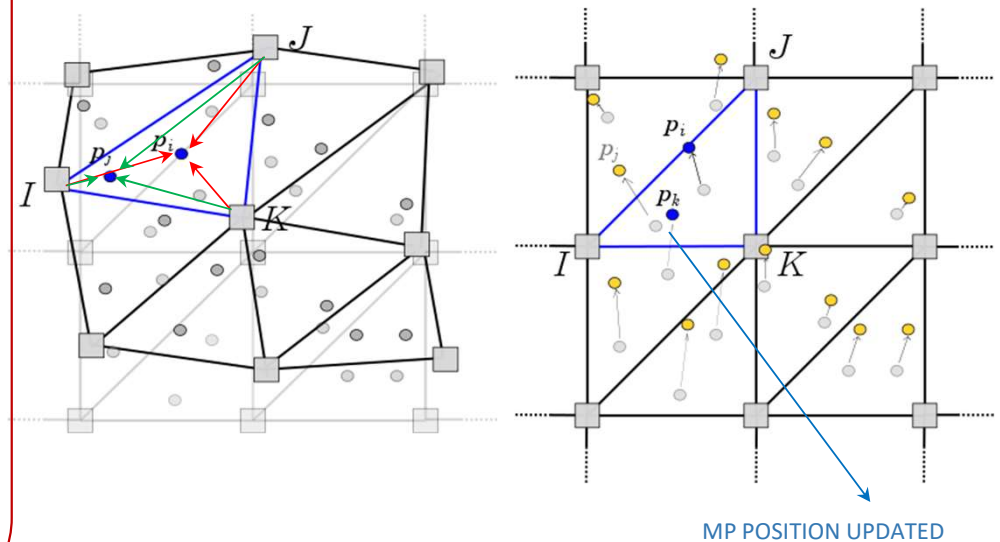
Mapping of the (time step) initial conditions on the FE grid's nodes

FEM U-L



UL FEM solution on the nodes of the background grid

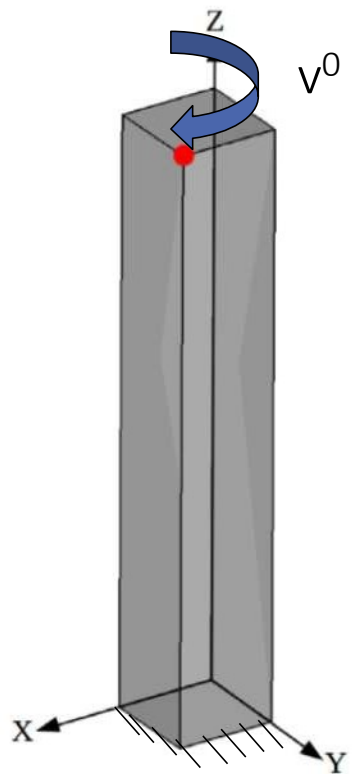
CONVECTIVE



Information is interpolated and stored on the particles which are moved on the calculated positions
 The grid is reset, and the MP connectivity is updated



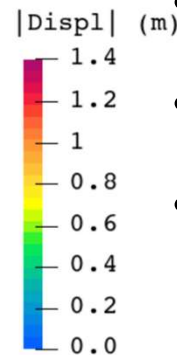
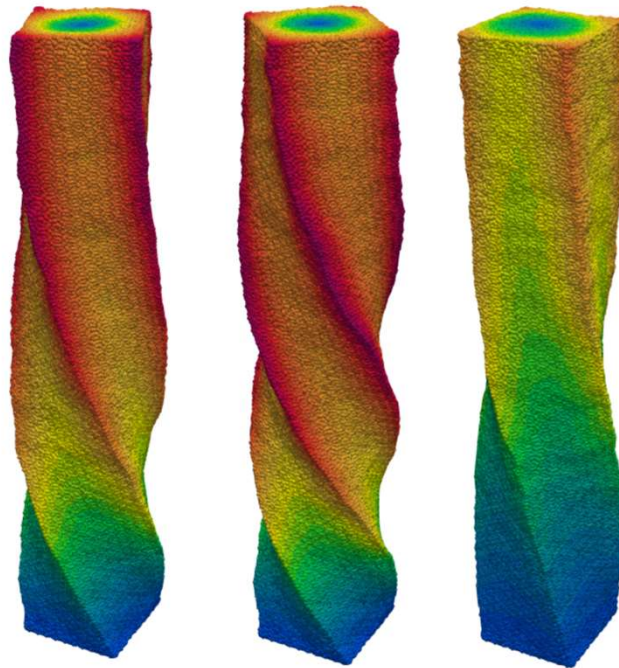
The key features of MPM.



0.05s

0.1s

0.2s



- Fixed background mesh ✓
- No remeshing ✓
- No front tracking technique ✓
- Continuum-based method ✓

Dt = 0.001s,
T_{fin}=0.5s
Mesh size h=0.2m
16 MPs/element

$$\mathbf{v}^0(x, y, z) = 105 \sin\left(\frac{\pi z}{12}\right) (y, -x, 0)^T \text{ m/s}$$

Twisting of a beam (large strain regime) using mixed U-P MPM stabilized with VMS.



Our research on MPM

- **MPM formulations**
 - Irreducible
 - Mixed U-P with stabilization
- Fluid (velocity-based) and solid (displacement-based) mechanics **formulations for granular flows**
- **Coupled Multiphysics Problems**
 - MPM-MPM
 - MPM-DEM
 - MPM-FEM



MPM (solid mechanics) formulations

- Irreducible formulation (U)
- Mixed formulation (U-P)



Formulations

Updated Lagrangian Formulation

Irreducible formulation

Unknown: $\mathbf{u}$

$$\begin{cases} \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} - \nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{b} & \text{in } \varphi(\Omega)(t), \\ \mathbf{u} = \bar{\mathbf{u}} & \text{on } \varphi(\partial\Omega_D)(t), \\ \boldsymbol{\sigma} \cdot \mathbf{n} = \bar{\mathbf{t}} & \text{on } \varphi(\partial\Omega_N)(t), \end{cases}$$

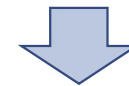


$$\mathbf{K}^{tan} \delta \mathbf{u} = -\mathbf{R}.$$

Mixed formulation

Unknowns: $\mathbf{u}, p$

$$\begin{cases} \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} - \nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{b} & \text{in } \varphi(\Omega)(t), \\ \frac{p}{\kappa} - \left(1 - \frac{1}{J}\right) = 0 & \text{in } \varphi(\Omega)(t), \\ \mathbf{u} = \bar{\mathbf{u}} & \text{on } \varphi(\partial\Omega_D)(t), \\ \boldsymbol{\sigma} \cdot \mathbf{n} = \bar{\mathbf{t}} & \text{on } \varphi(\partial\Omega_N)(t), \end{cases}$$



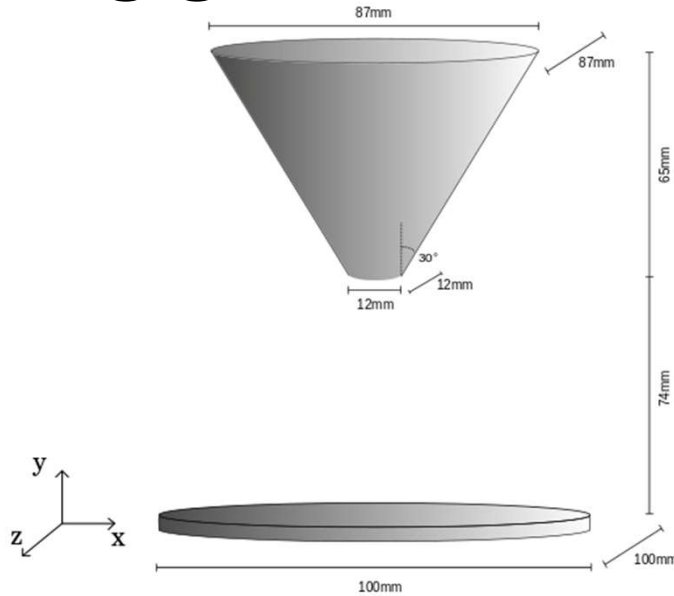
$$\tilde{\mathbf{K}}^{tan} \cdot \delta \mathbf{U} = \begin{pmatrix} \mathbf{K}^{tan} & \mathbf{B}^t \\ \mathbf{B} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \delta \mathbf{u} \\ \delta p \end{pmatrix} = \begin{pmatrix} \mathbf{R}^u \\ \mathbf{R}^p \end{pmatrix}$$



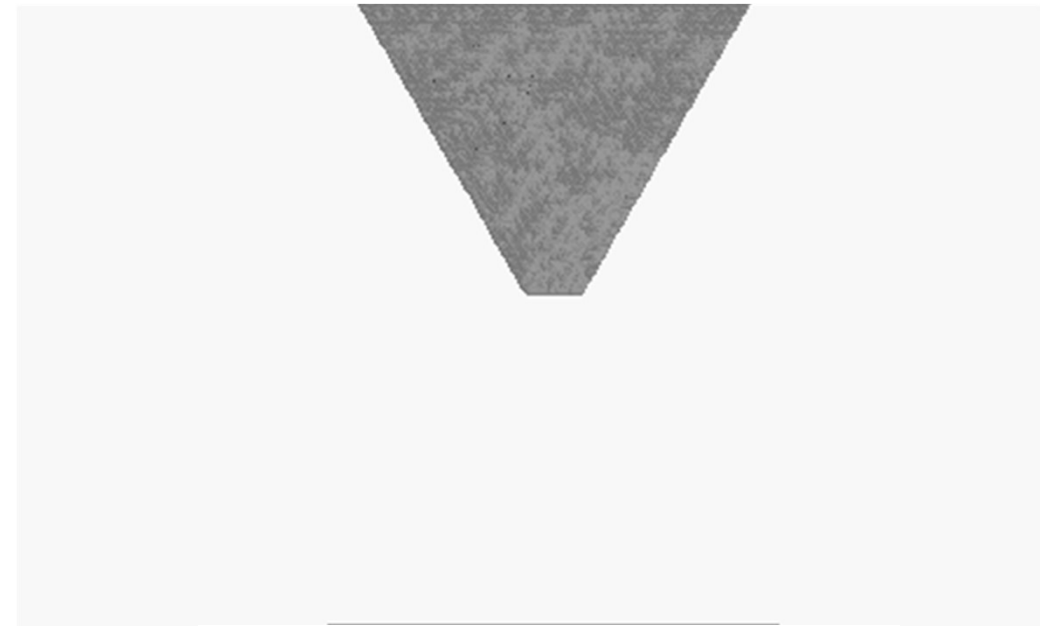
Modeling granular flows



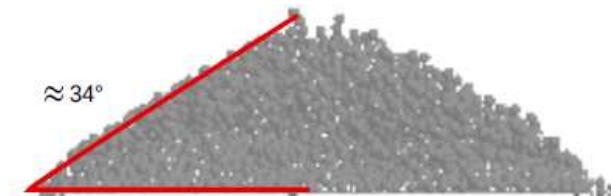
 Research and Development



REPOSE ANGLE TEST



Proprietary information of Nestlé S.A., Vevey, Switzerland

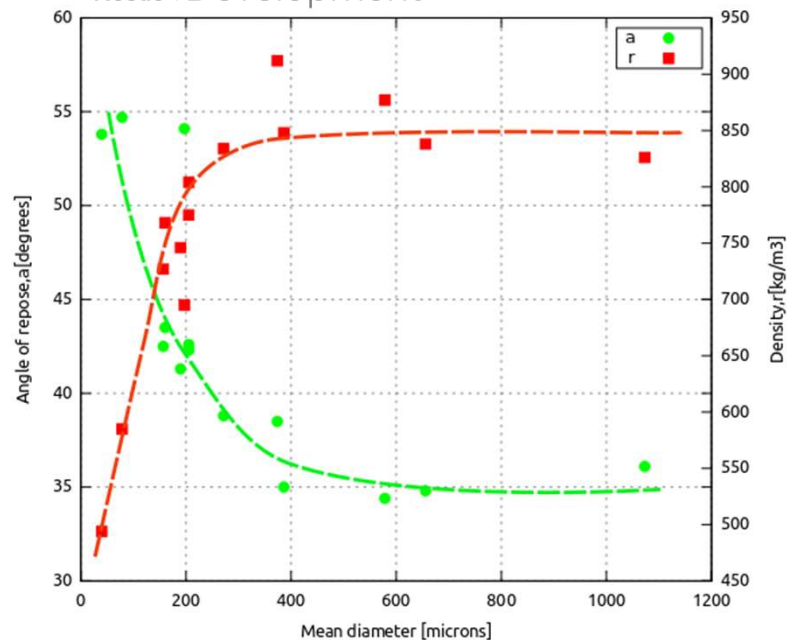


Modeling granular flows



GRANUFALL TEST

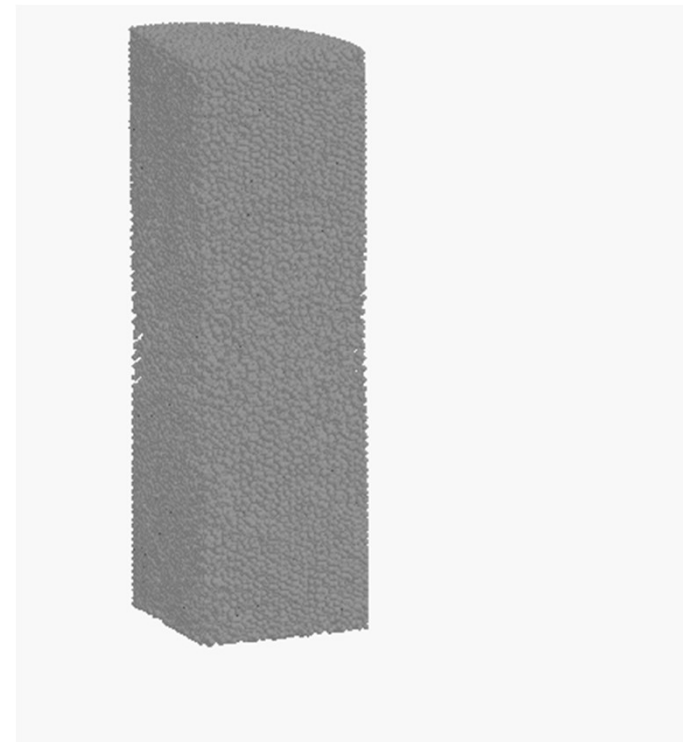
CASE a $\phi = 18$ mm



Proprietary information of Nestlé S.A., Vevey, Switzerland

a: Carr index (compressibility)

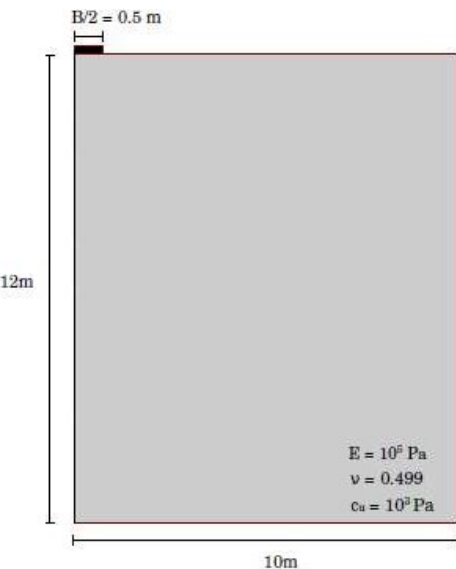
r: density



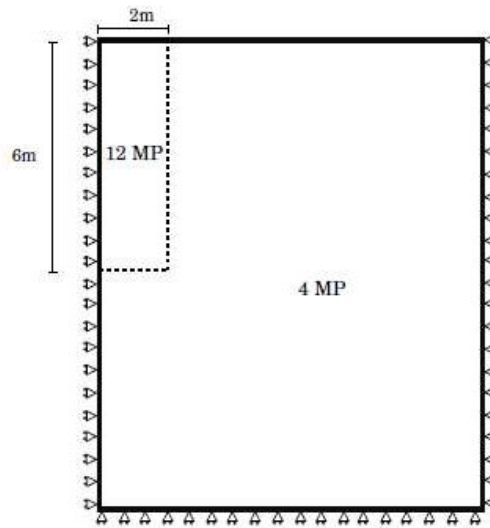
VOLUMETRIC FLOW [mL/s] = 63.07



The Rigid Strip Footing test

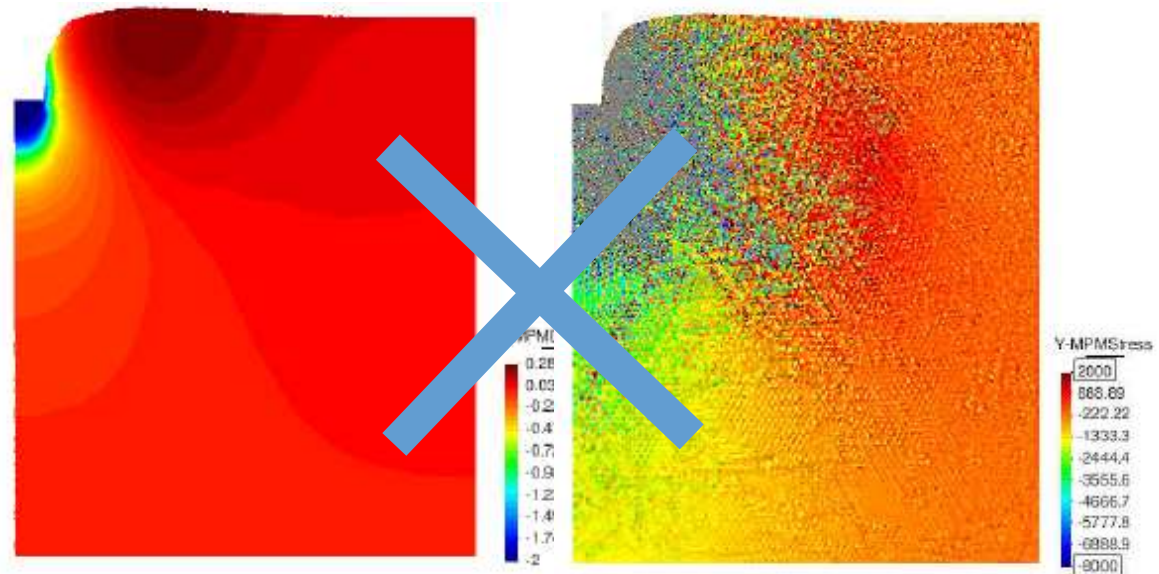


TRESCA MODEL IN
FINITE STRAINS



[Nazem et al., 2006]

A classical irreducible formulation fails



Vertical
displacement

Vertical Cauchy
stress



Constitutive laws

SMALL DEFORMATION REGIME - Linear elasticity:

$$\boldsymbol{\sigma}^{\text{dev}} = \mathbb{C}^{\text{dev}} : \nabla^s \mathbf{u}$$

LARGE DEFORMATION REGIME -

$$\boldsymbol{\sigma} = \frac{1}{J} \mathbf{F} \mathbf{S} \mathbf{F}^T \longrightarrow \mathbf{S} := J \mathbf{F}^{-1} \boldsymbol{\sigma}^{\text{dev}} \mathbf{F}^{-T} + p J \mathbf{C}^{-1}$$

- Hyperelasticity: $\mathbf{S} := 2 \frac{\partial \Psi(\mathbf{C})}{\partial \mathbf{C}} = 2 \frac{\partial \Psi_{\text{dev}}}{\partial \mathbf{C}} + \frac{d\Psi_{\text{vol}}}{dJ} J \mathbf{C}^{-1}$

Neo-Hookean:

$$\Psi_{\text{dev}}(\bar{I}_1) = \frac{\mu}{2} (\bar{I}_1 - 3)$$

$$\Psi_{\text{vol}}(J) = \kappa (J - \log(J) - 1)$$

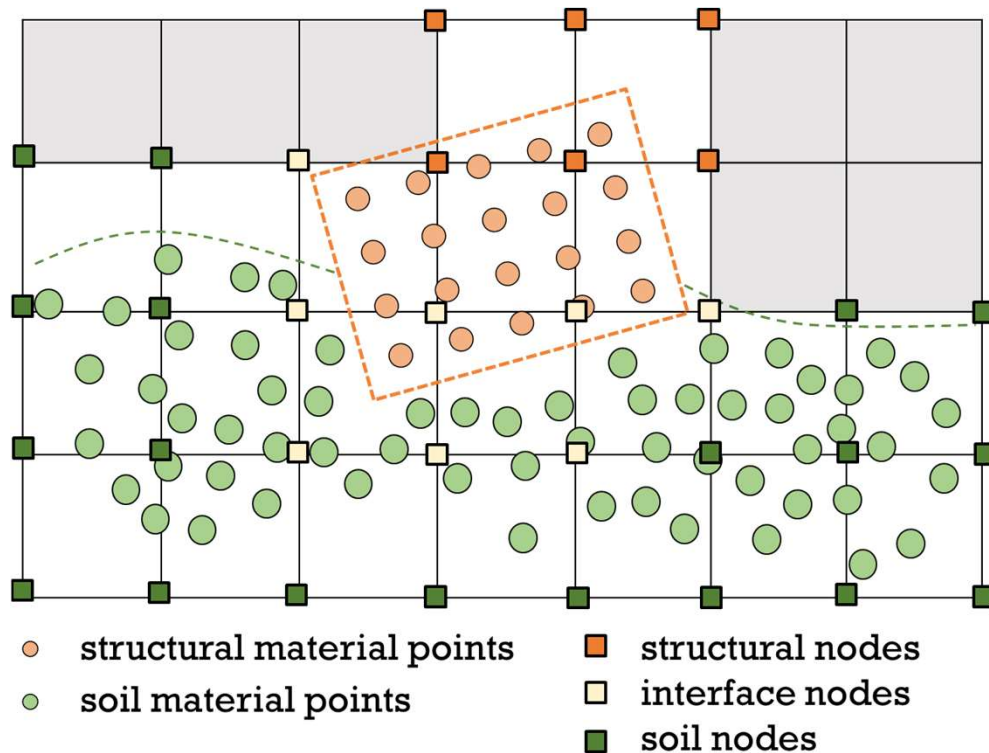
- Elastoplasticity (Mohr Coulomb, Ducker Preager,...)
- Viscoplasticity (Newtonian law): $\boldsymbol{\sigma}^{\text{dev}} = 2\mu \nabla^s \mathbf{v}$



Multiphysics simulations: MPM-MPM “natural” coupling



MPM-MPM “monolithic coupling”



Solve both the structure and soil together:

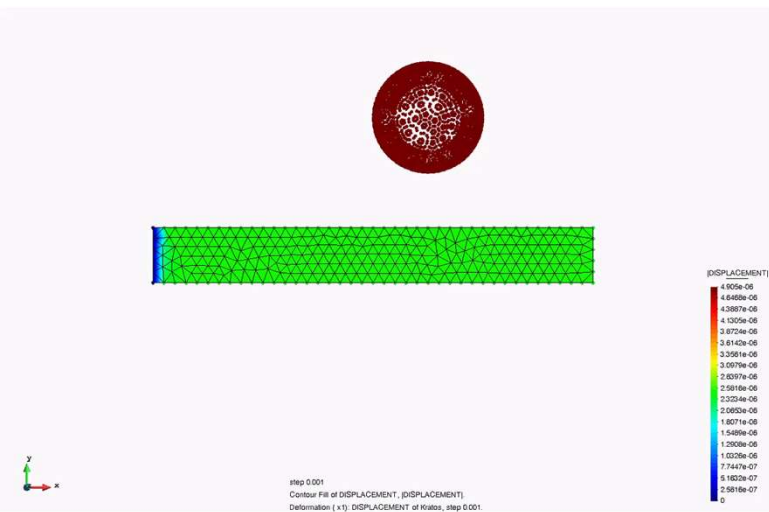
$$\begin{bmatrix} \mathbf{K}_s & \mathbf{K}_{s\Gamma} & 0 \\ \mathbf{K}_{\Gamma s} & \mathbf{K}_{\Gamma} & \mathbf{K}_{\Gamma st} \\ 0 & \mathbf{K}_{st\Gamma} & \mathbf{K}_{st} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u}_s \\ \Delta \mathbf{u}_{\Gamma} \\ \Delta \mathbf{u}_{st} \end{bmatrix} = - \begin{bmatrix} \mathbf{R}_s \\ \mathbf{R}_{\Gamma} \\ \mathbf{R}_{st} \end{bmatrix} .$$

Stable, robust, but producing sticking contact behavior

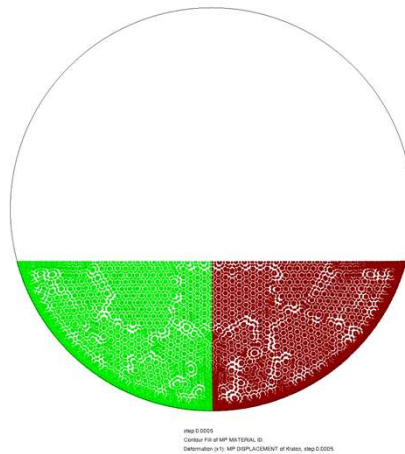


MPM-MPM

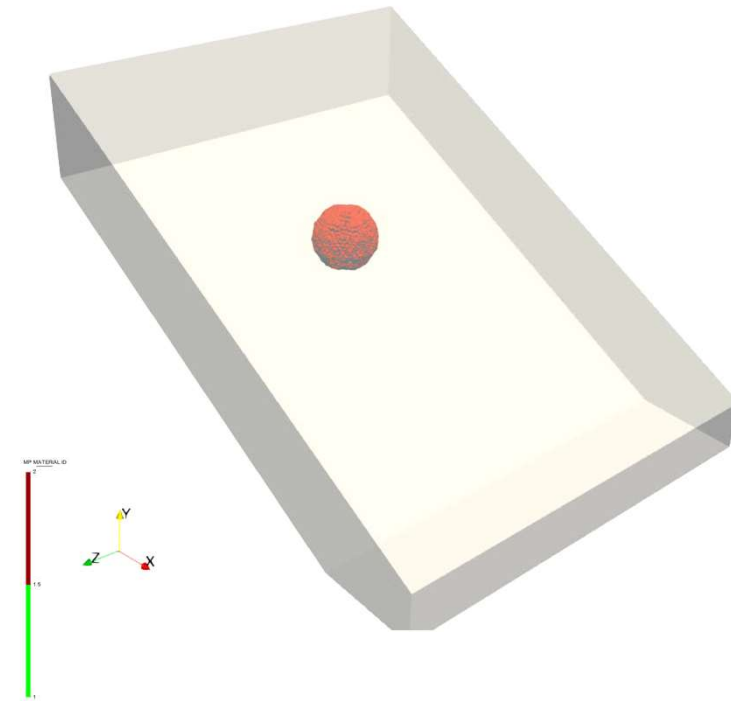
- Heterogeneous materials
- No penetration condition
- BUT
- Sticky contact



Granular ball impacting on a bending beam



Mixer of two granular materials



Multi-material ball
half fluid (blue) - half clay (red)



Multiphysics simulations.

Partitioned coupling

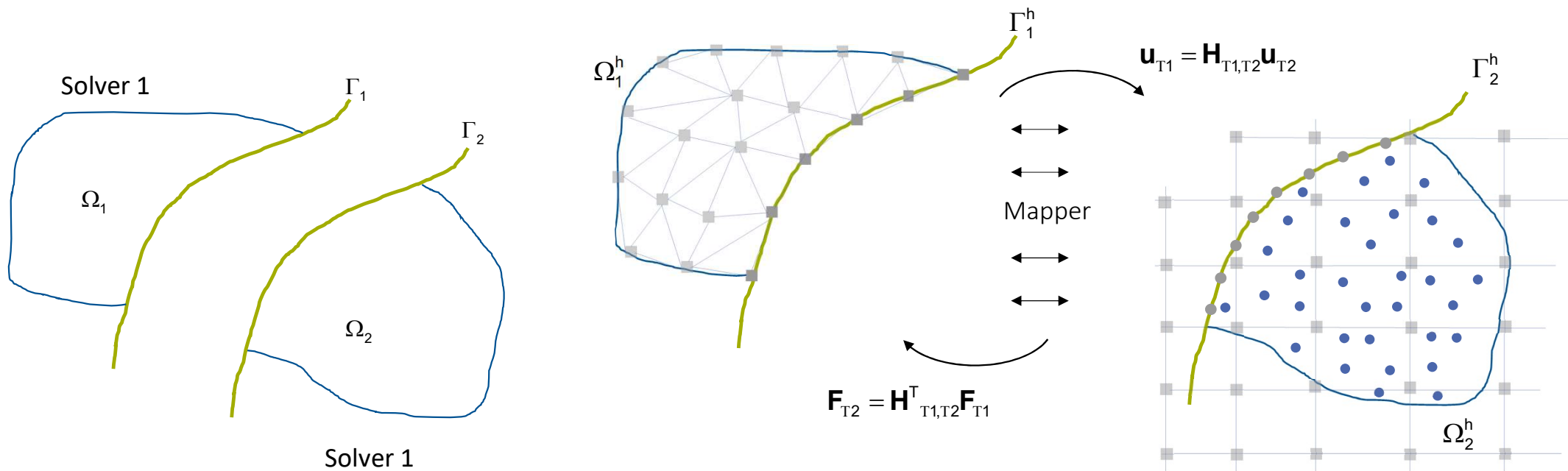
- MPM-DEM
- MPM-FEM



PARTITIONED COUPLING

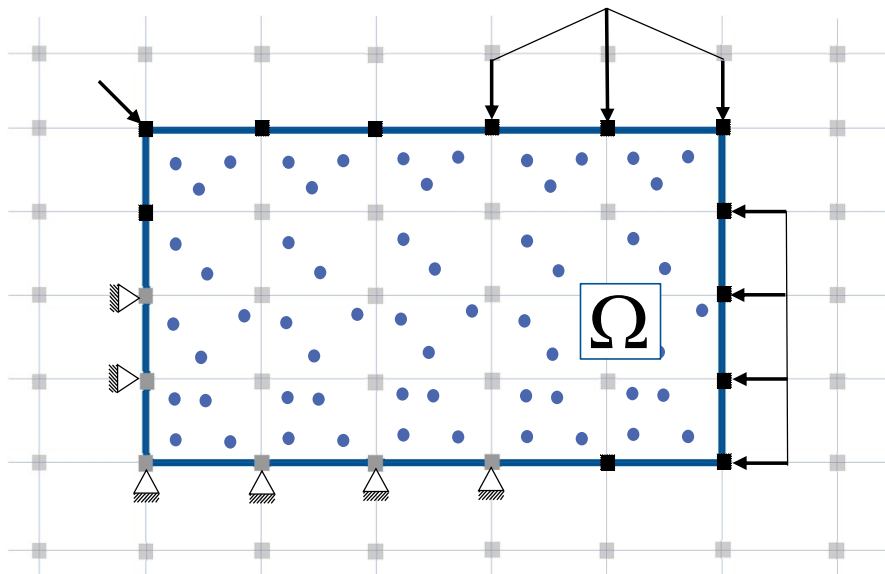
Black-box solver

Data Mapping



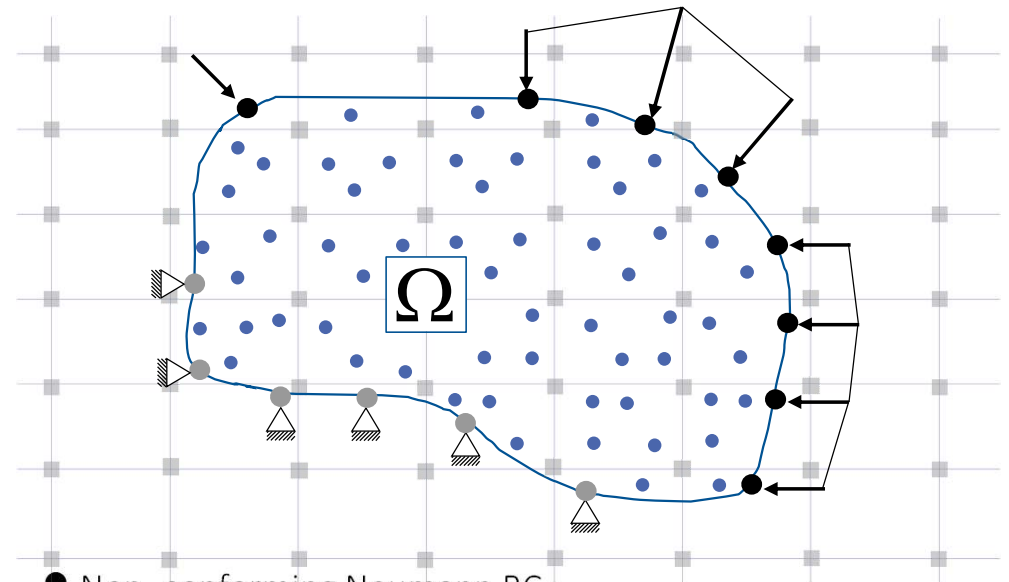
Boundary Conditions in MPM

CONFORMING: Imposition at nodes of background grid



- Grid conforming Dirichlet Boundary Condition
- Grid conforming Neumann Boundary Condition

NON CONFORMING BCs: Boundary Particles are introduced

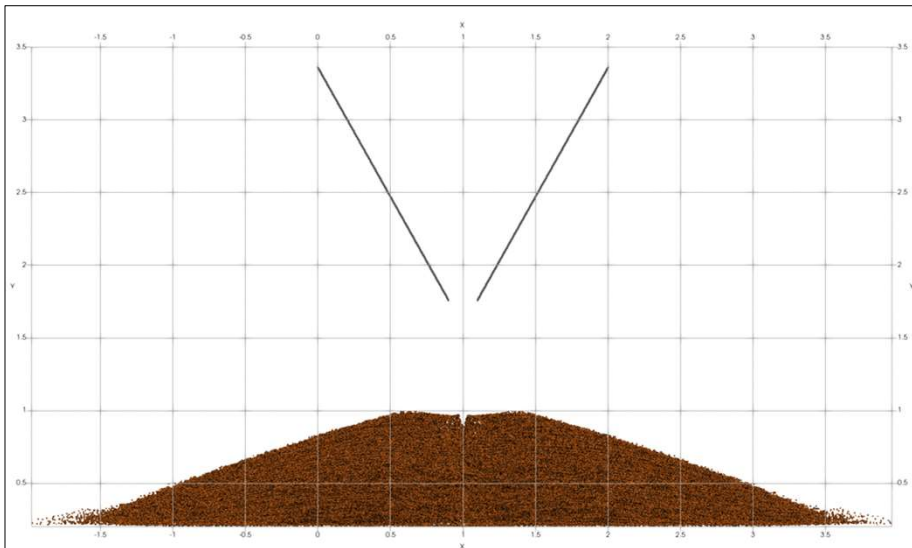


- Non- conforming Neumann BC
 - ✓ Point Load Condition
 - ✓ Line Load Condition
- Non- conforming Dirichlet BC (fixed, slip, contact)



NON-CONFORMING BOUNDARY CONDITIONS in MPM

Boundary Particles are introduced



Non- conforming Neumann BC

- ✓ Point Load Condition
- ✓ Line Load Condition

Non- conforming Dirichlet BC (fixed, slip, contact)

- ✓ Penalty Imposition
- ✓ Lagrange Multiplier Imposition

Penalty Method

$$\delta W^{\text{Penalty}} = \beta \int_{\Gamma_D} (\mathbf{u} - \bar{\mathbf{u}}) \cdot \delta \mathbf{u} d\Gamma_D$$

Lagrange Multiplier Method

$$\delta W^{\text{LM}}(\mathbf{u}, \boldsymbol{\lambda}, \delta \mathbf{u}) = \int_{\Gamma_D} \boldsymbol{\lambda}^T \delta \mathbf{u} d\Gamma_D$$

$$\delta W^{\text{LM}}(\mathbf{u}, \boldsymbol{\lambda}, \delta \boldsymbol{\lambda}) = \int_{\Gamma_D} \delta \boldsymbol{\lambda}^T (\mathbf{u} - \bar{\mathbf{u}}) d\Gamma_D$$



MPM-DEM

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MPM APPLICATION - Kratos Workshop 2024 - Padova

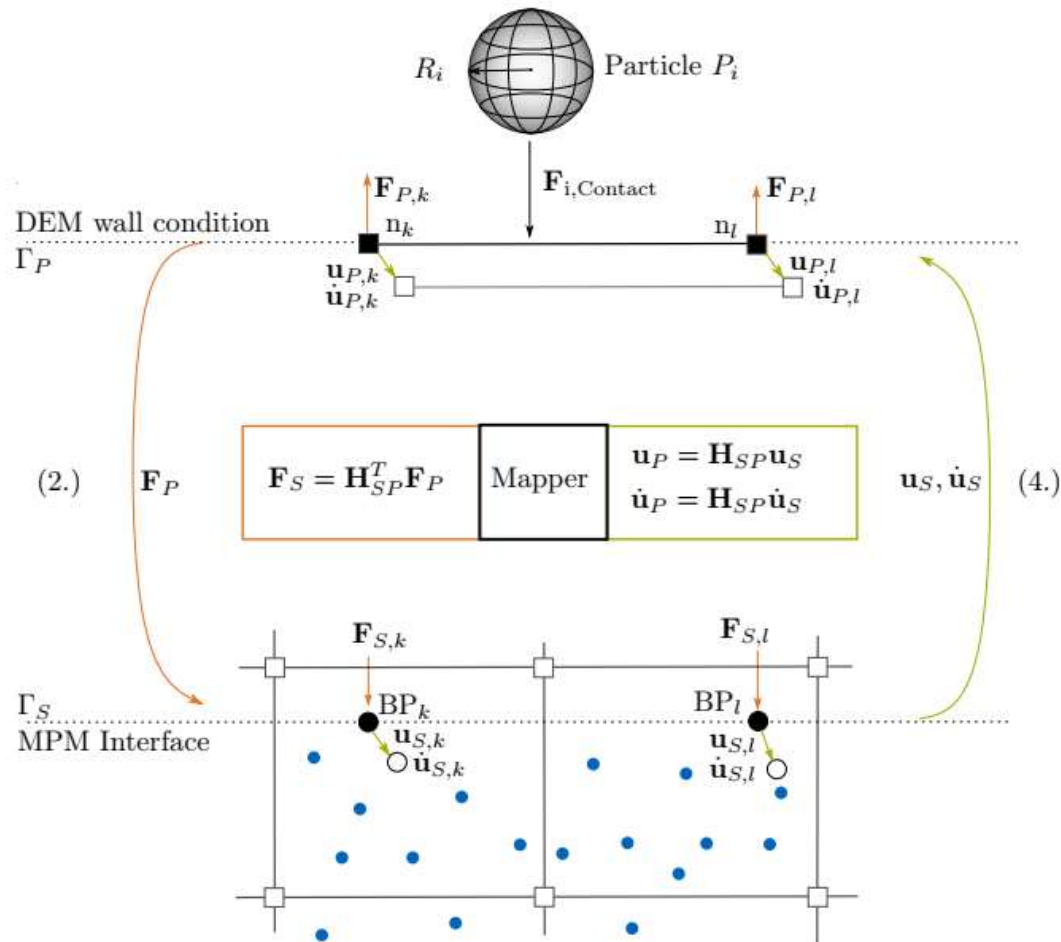
32



Partitioned MPM-DEM

(1.) DEM solver:

$$\mathbf{F}_P = \text{DEM}(\mathbf{u}_P, \dot{\mathbf{u}}_P)$$



- Material points
- Boundary particles BP

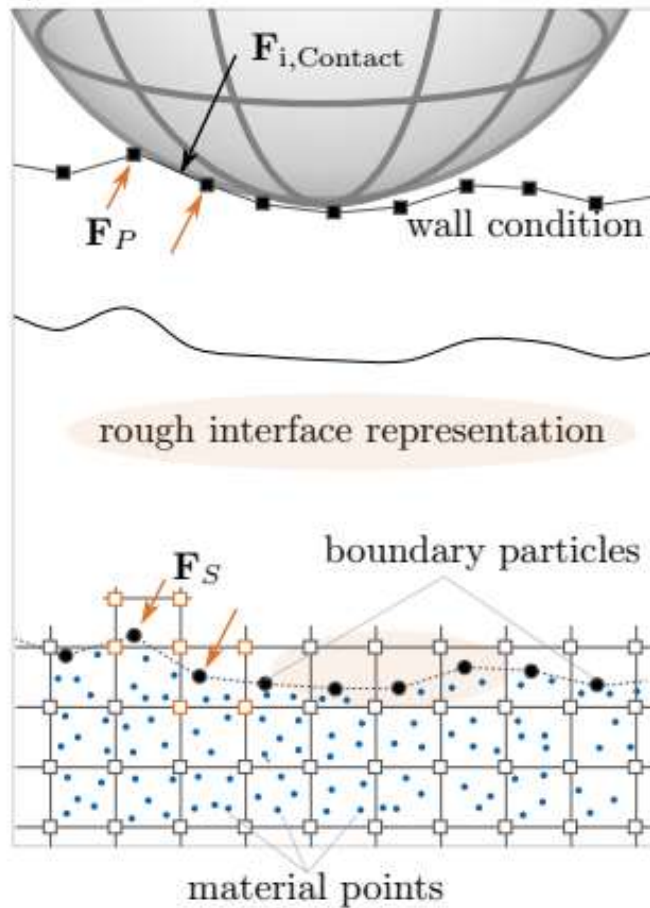
(3.) MPM solver:

$$\mathbf{u}_S, \dot{\mathbf{u}}_S = \text{MPM}(\mathbf{F}_S)$$

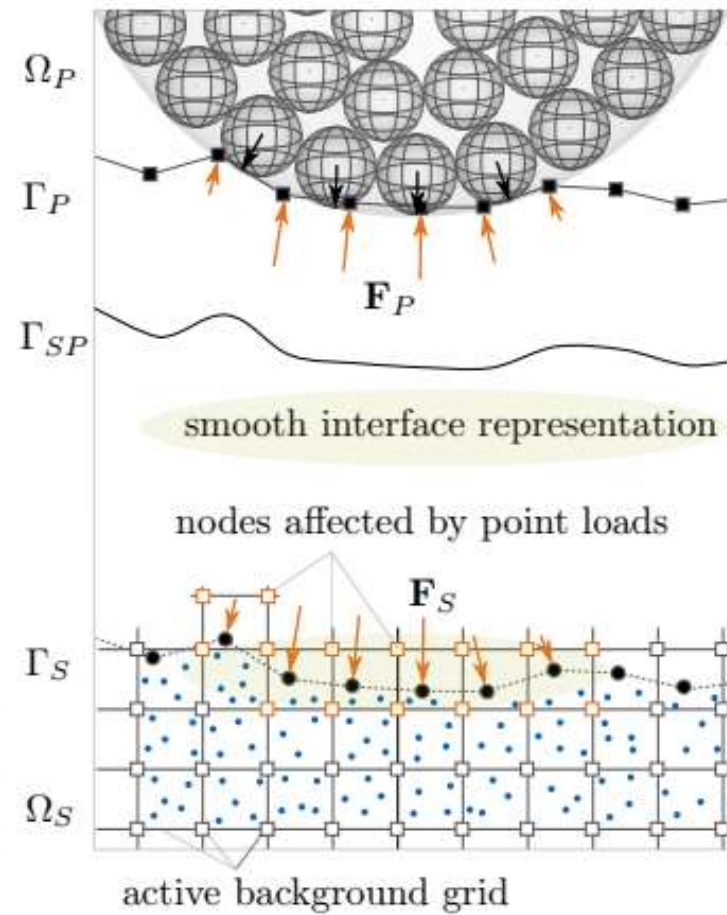


MPM-DEM coupling

a) DEM Particle

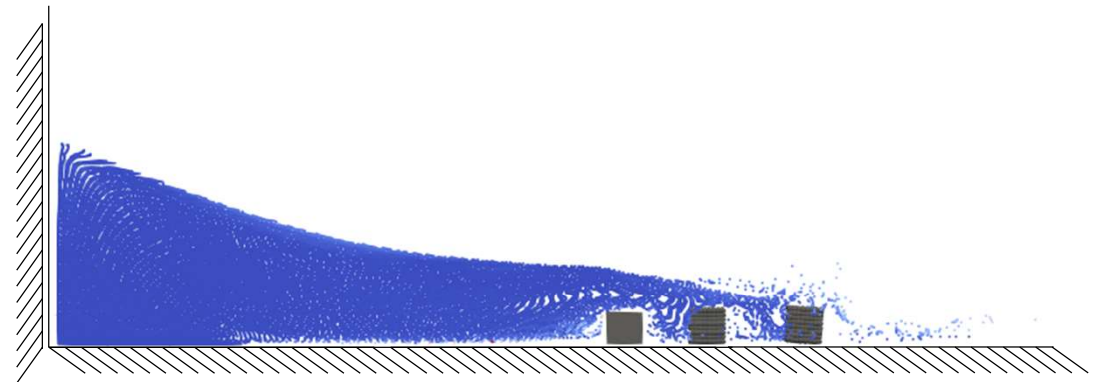
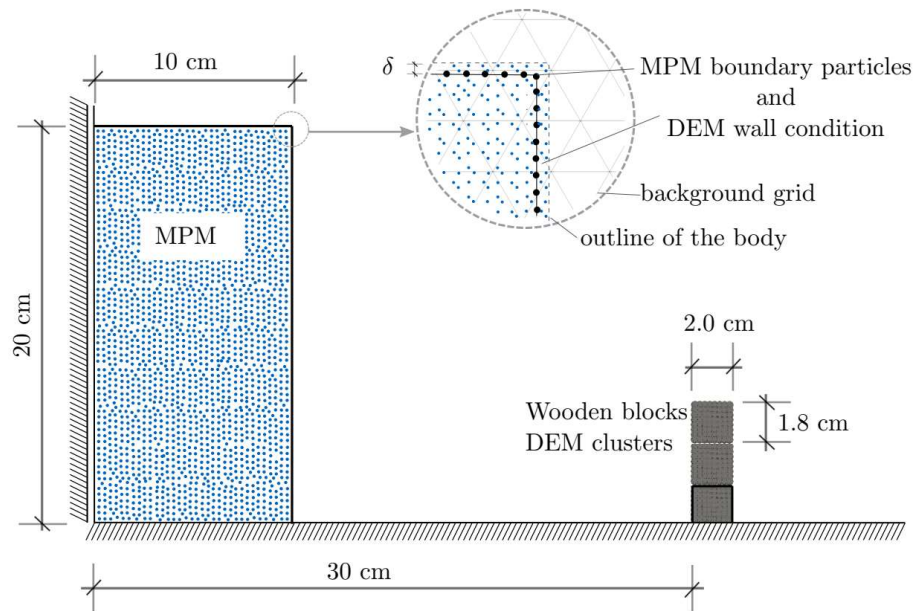


b) DEM Particle Clustering



PARTITIONED MPM-DEM COUPLING

Experiment: Granular Flow impacting discrete particles

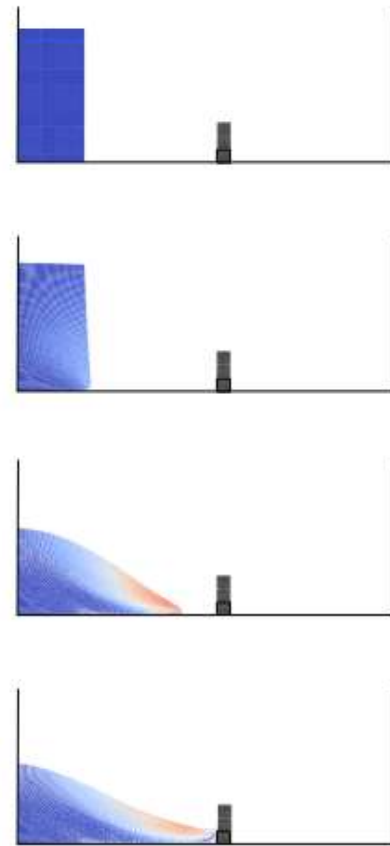
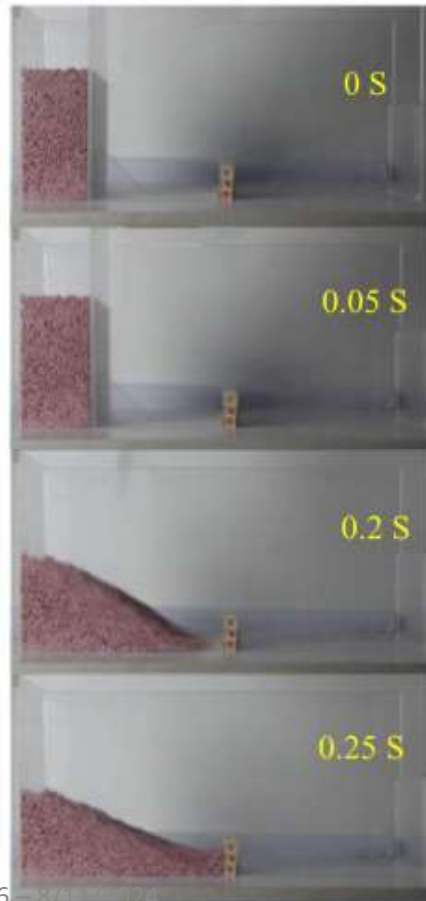


Singer, V., Sautter, K.B., Larese, A. *et al.* A partitioned material point method and discrete element method coupling scheme. *Adv. Model. and Simul. in Eng. Sci.* **9**, 16 (2022).

PARTITIONED MPM-DEM COUPLING

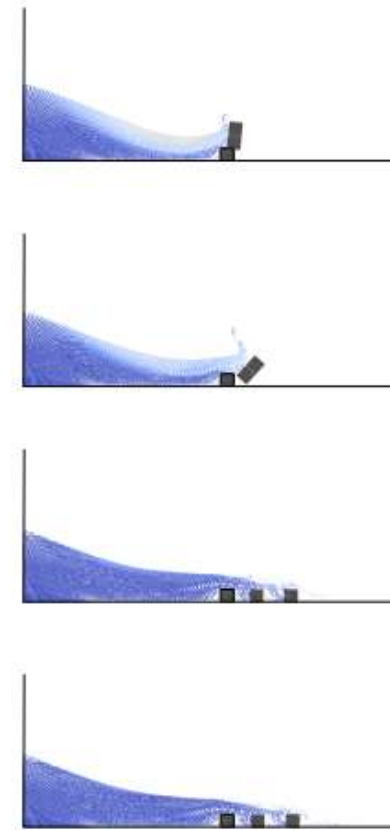
I) Experiment [36]

II) Simulation



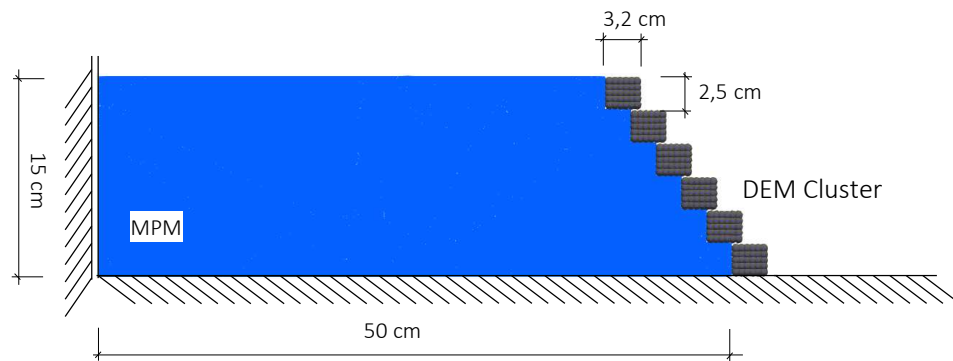
I) Experiment [36]

II) Simulation

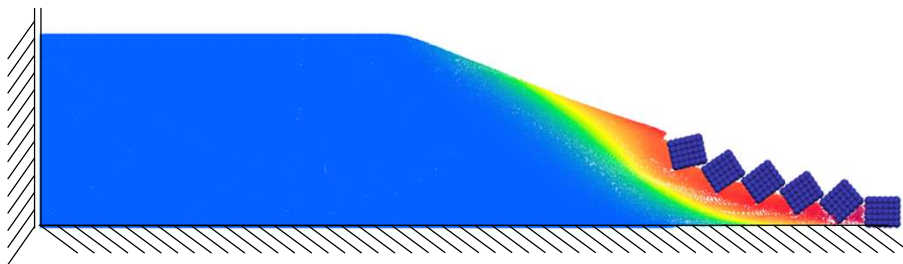


Partitioned MPM - DEM Coupling

Experiment: Retaining wall collapse



Simulation: MPM-DEM coupling

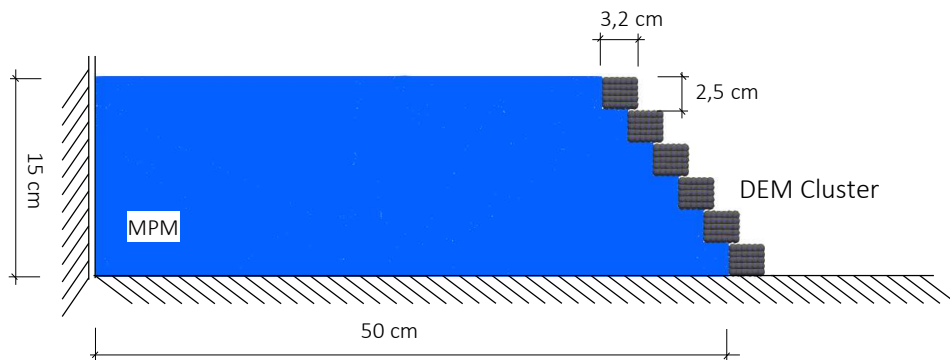


Bui, H. et al. (2014), *Int. J. Numer. Anal. Meth. Geomech.*, 38, 1321– 1340



Partitioned MPM - DEM Coupling

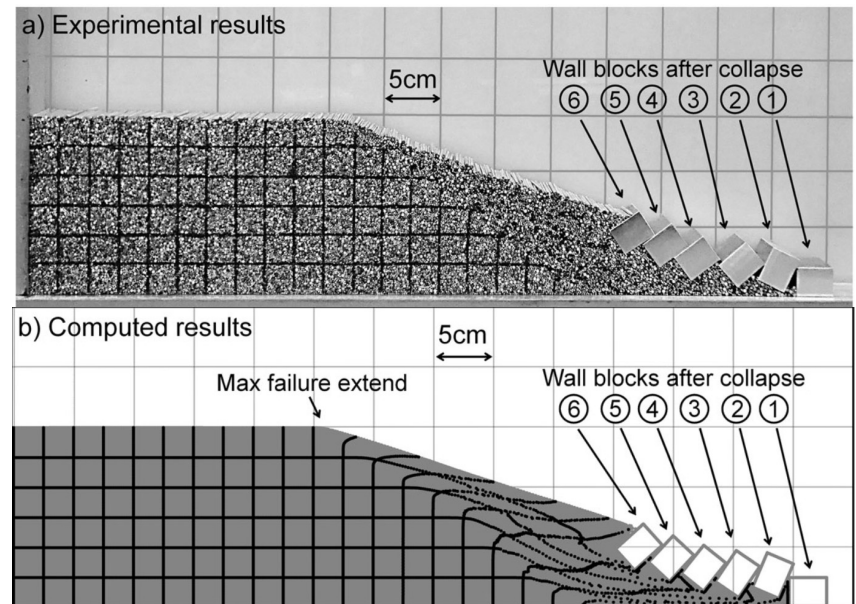
Experiment: Retaining wall collapse



Simulation: MPM-DM coupling



Literature:



Bui, H. et al. (2014), *Int. J. Numer. Anal. Meth. Geomech.*, 38, 1321– 1340



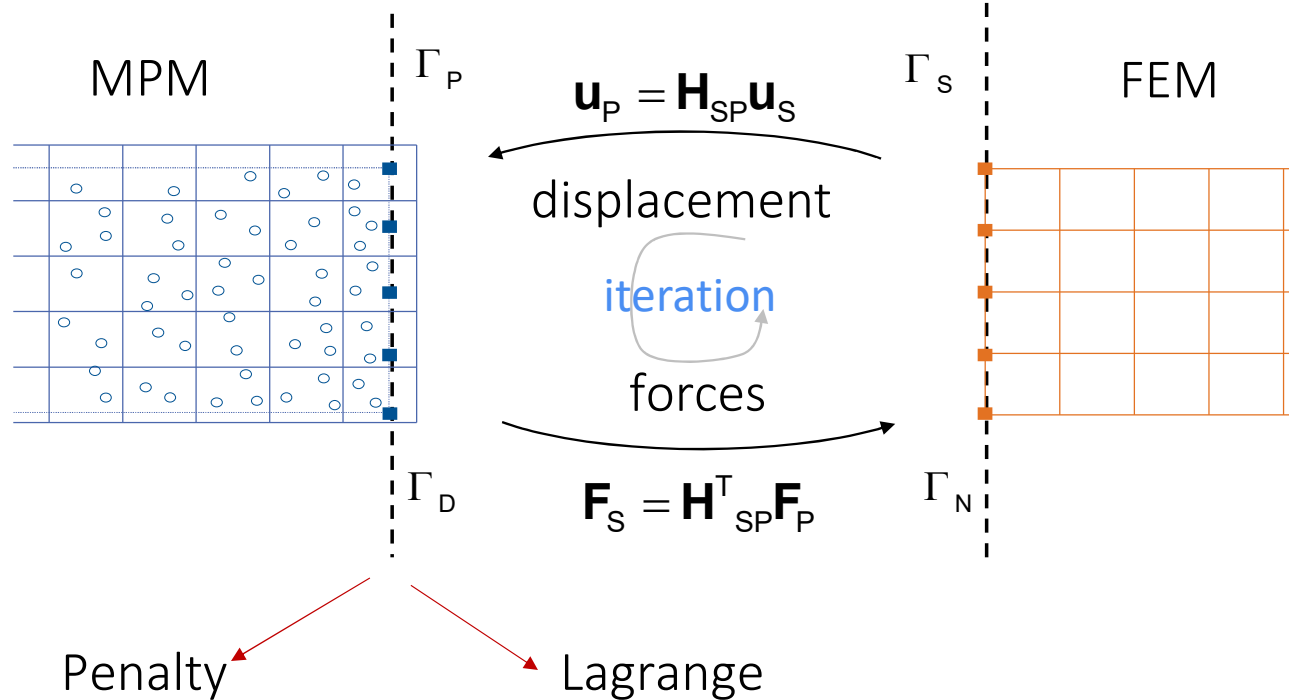
MPM-FEM



PARTITIONED MPM – FEM

Gauss Seidel Communication Pattern

Mapper

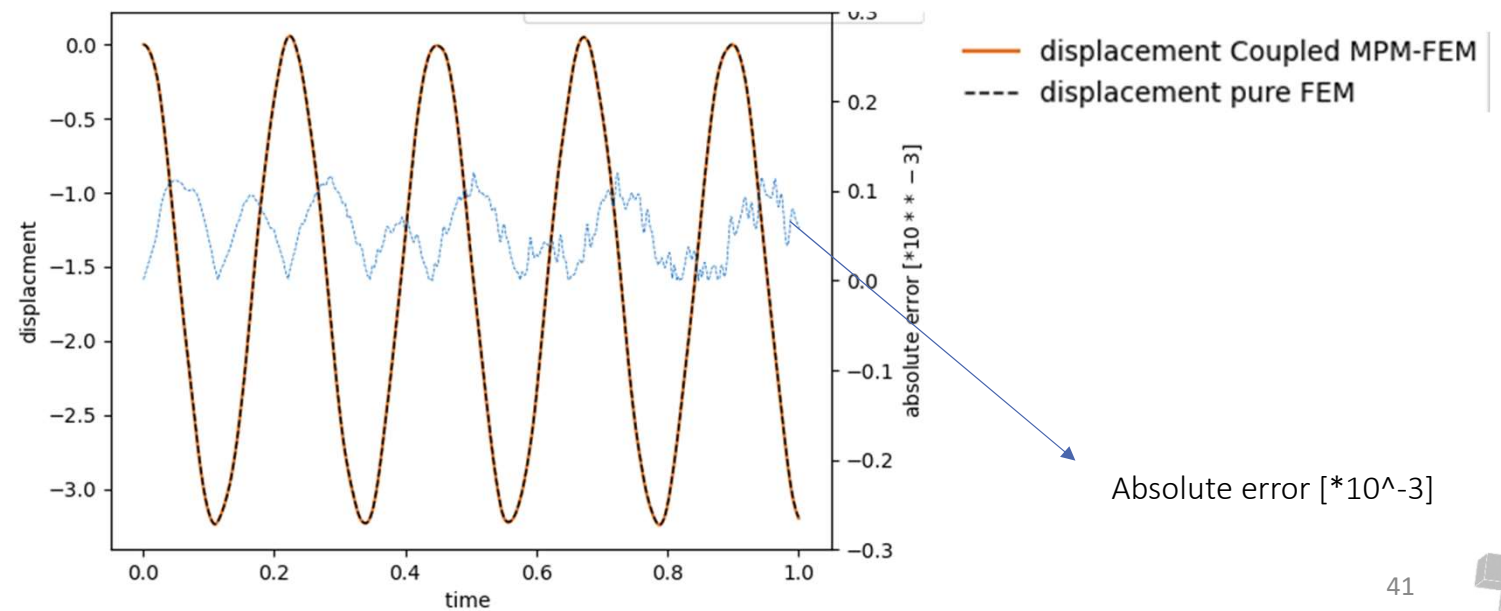
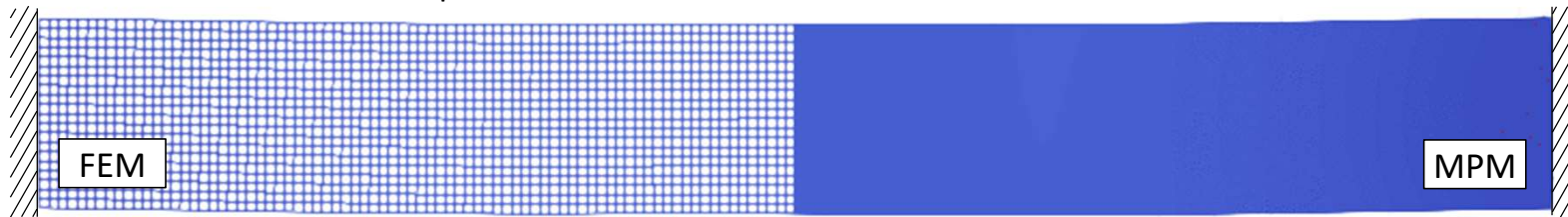


Chandra, B.; et al. Acta Geotechnica, 16(8), 2315-2335 (2021)



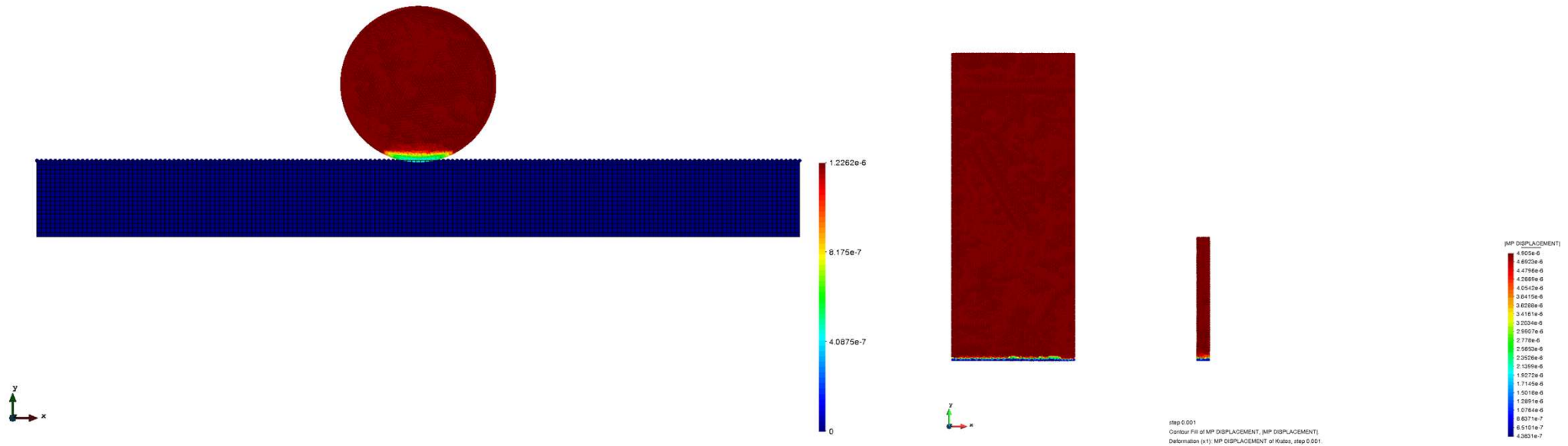
PARTITIONED MPM – FEM

Validation: Dynamic both sided clamped beam



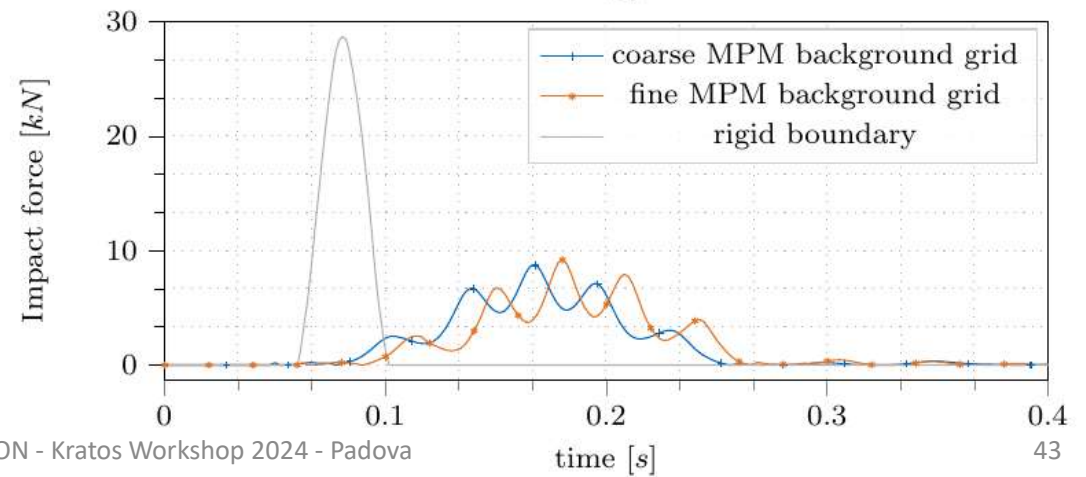
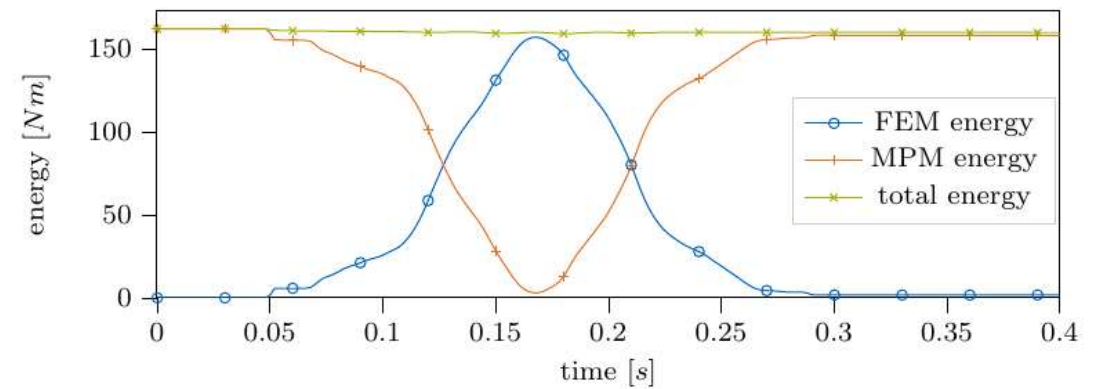
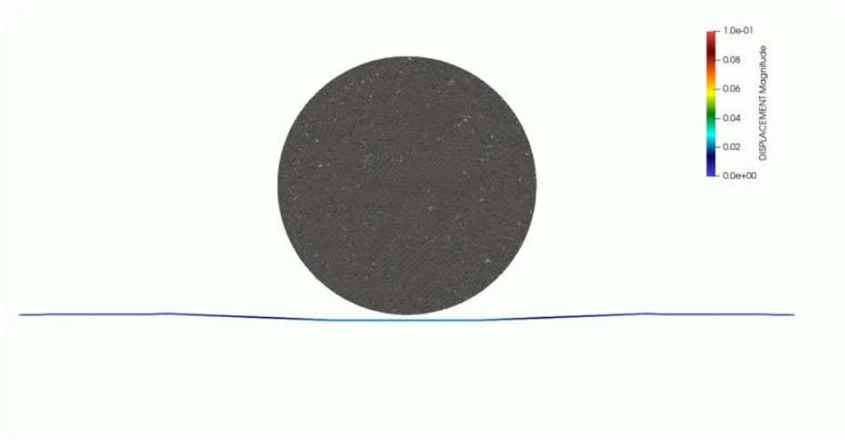
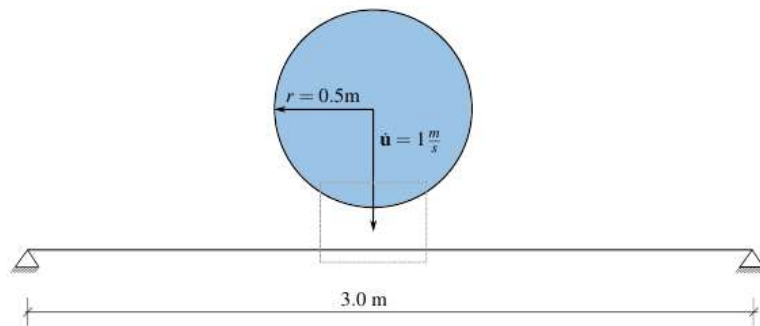
PARTITIONED MPM – FEM

...with granular material



PARTITIONED MPM – FEM

Impact MPM *elastic body* impacting a flexible structure



PARTITIONED MPM – FEM

Impact *granular material* flexible structure

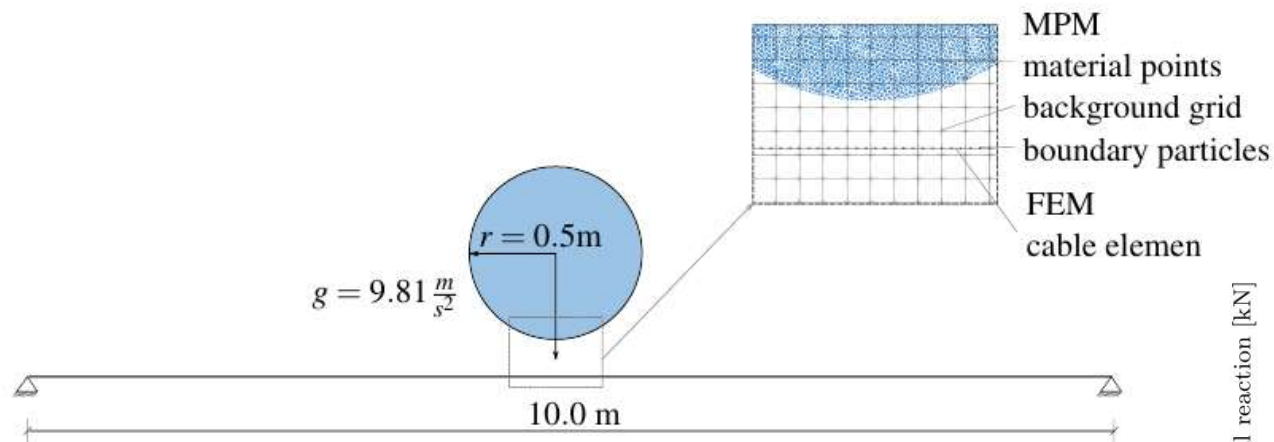


Figure 8.28: System of the cable and the cylinder made of granular material.

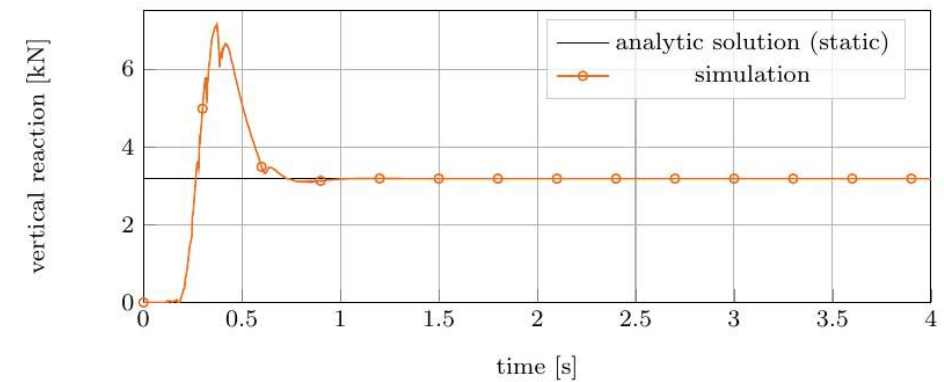


Figure 8.31: Reaction forces at the cable structure due to the impact of the granular material.

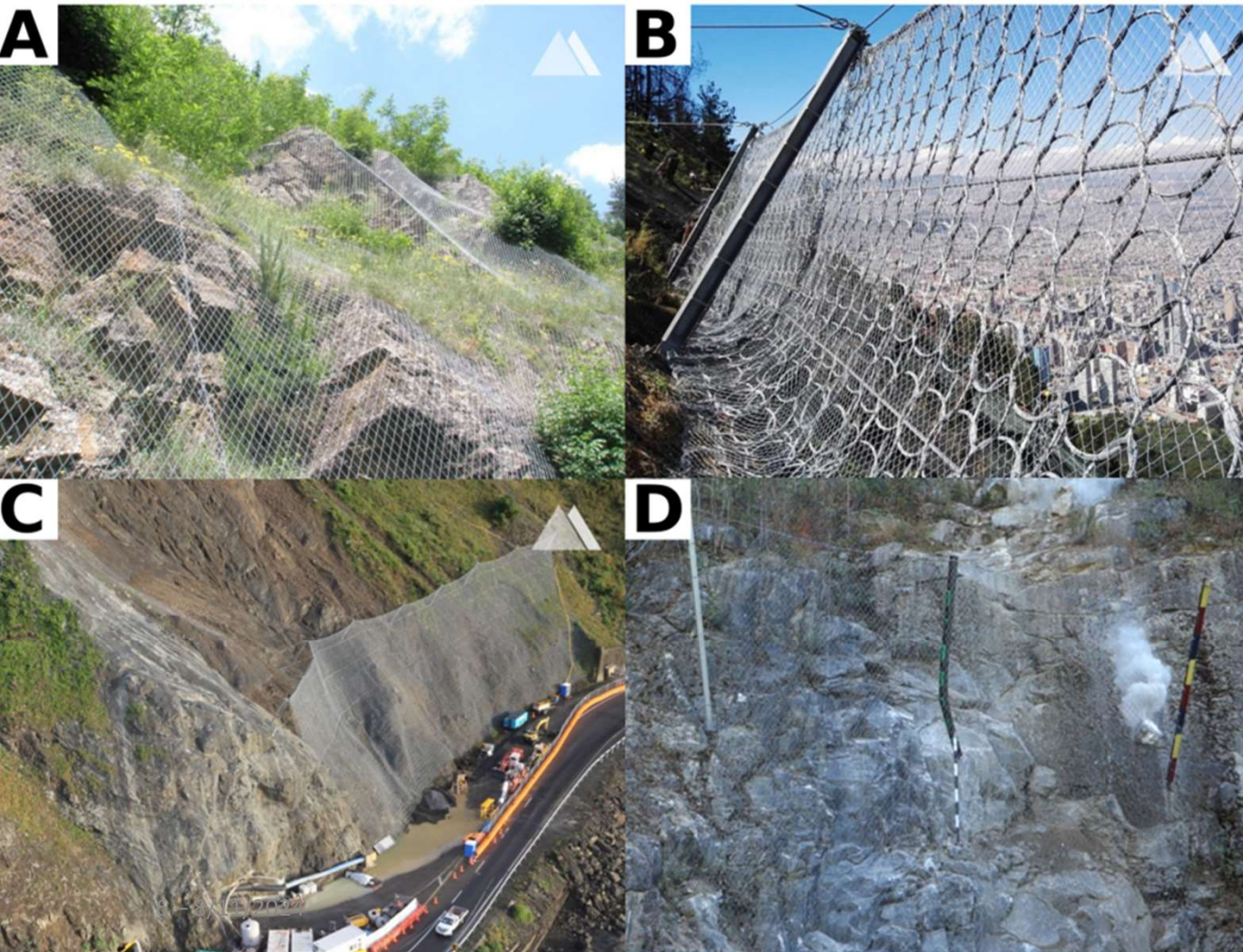


Our ongoing challenge

SIMULATION OF PROTECTIVE NETS



Protection systems



GEOBRUGG®
BRUGG
Safety is our nature

- A) Rockfall protection (diamond net)
- B) Mudflow protection (RING NET)
- C) Attenuator barrier
- D) Attenuator barrier



Rock fall –Protection

Structural net – Sliding Cable

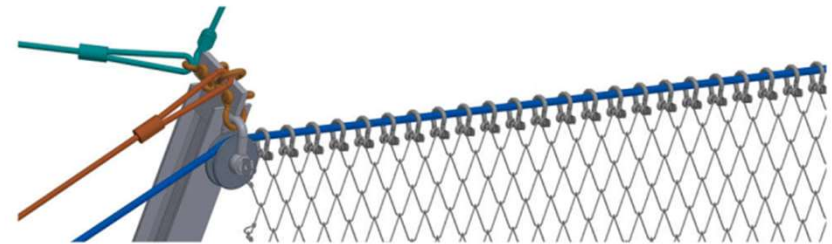
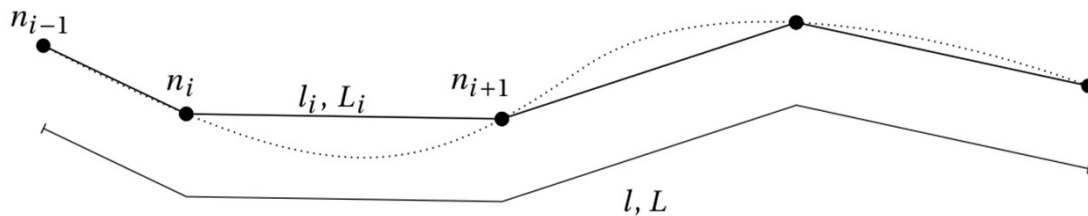


Figure 3.7: Sketch of upper rope installation, taken from [51] and adapted from [102], compare with Figure 3.8a. The blue line depicts the structural part which is modeled by the sliding cable element.



(a) Edge Installation.



(b) Upper rope installation.

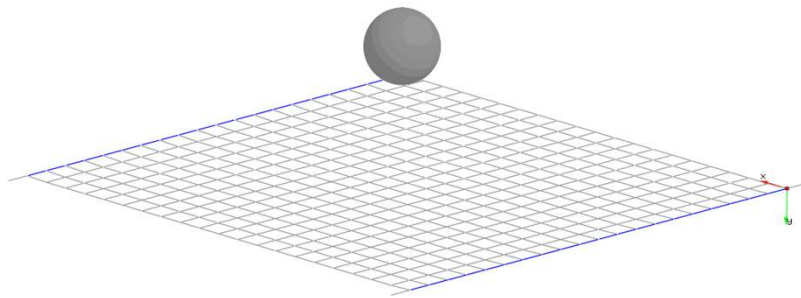
Figure 3.8: Photos of the real world model of the sliding cable element, taken in Walenstadt, Switzerland.



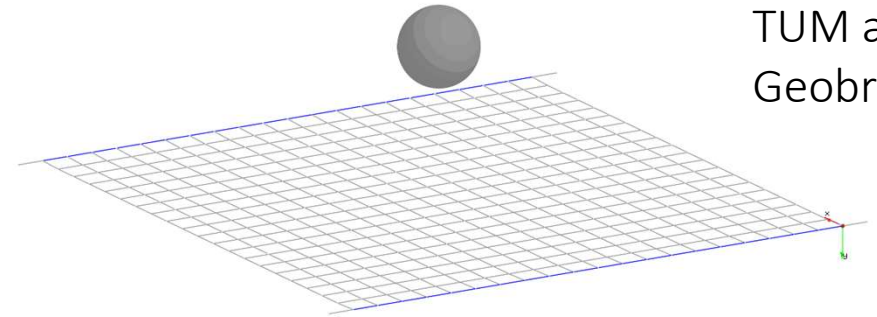
Rock fall –Protection

Structural net – Sliding Cable

Curtesy of
Klaus
Sautter,
TUM and
Geobrugg



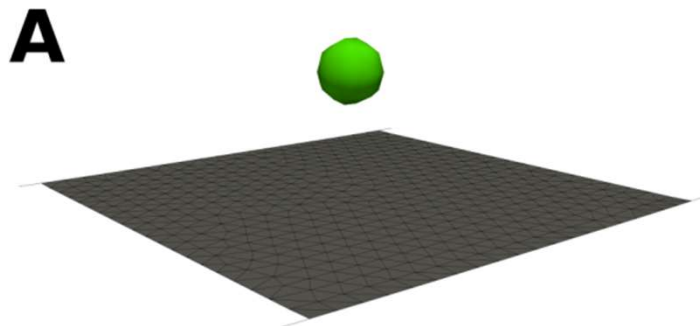
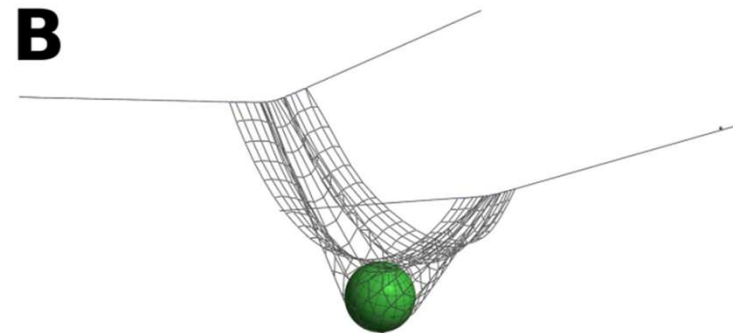
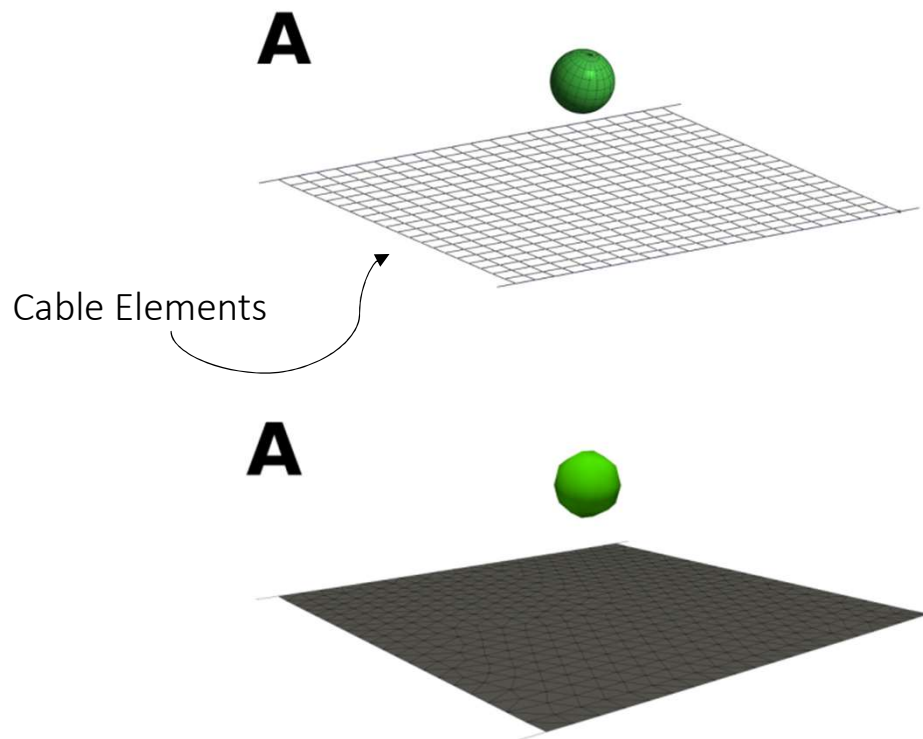
Including friction



Excluding friction



Rock fall –Protection



Courtesy of
Klaus
Sautter,
TUM and
Geobrugg

Membrane Elements with „tension field theory“
(excluding compression stresses)
Nakashino & Natori

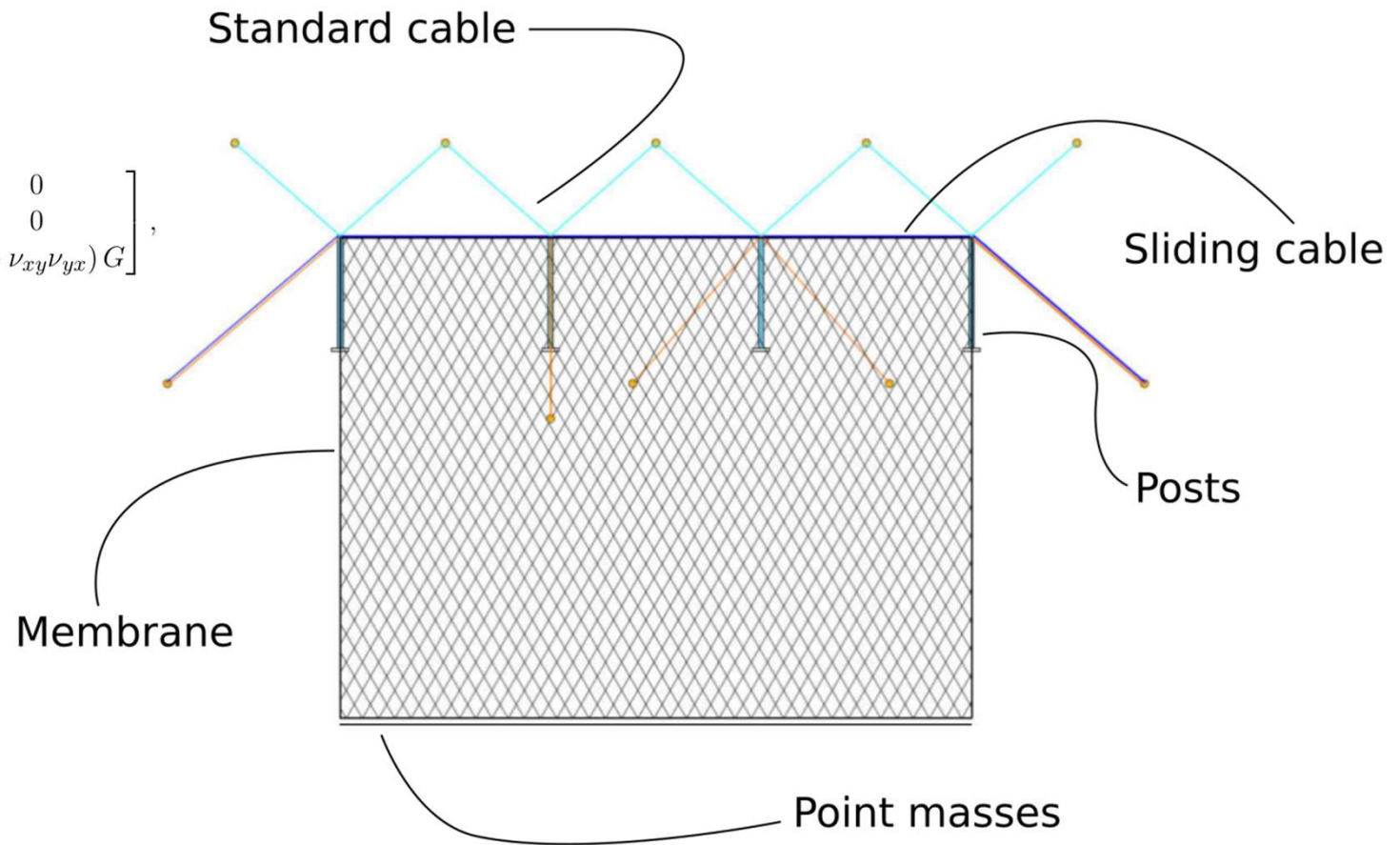


Attenuator

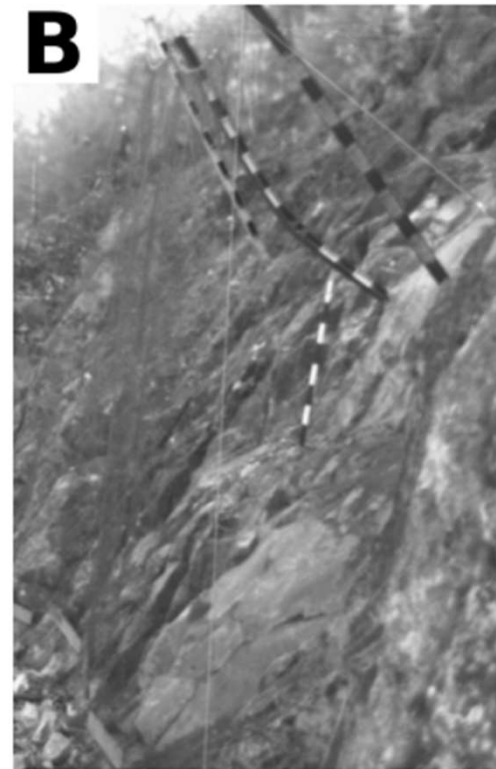
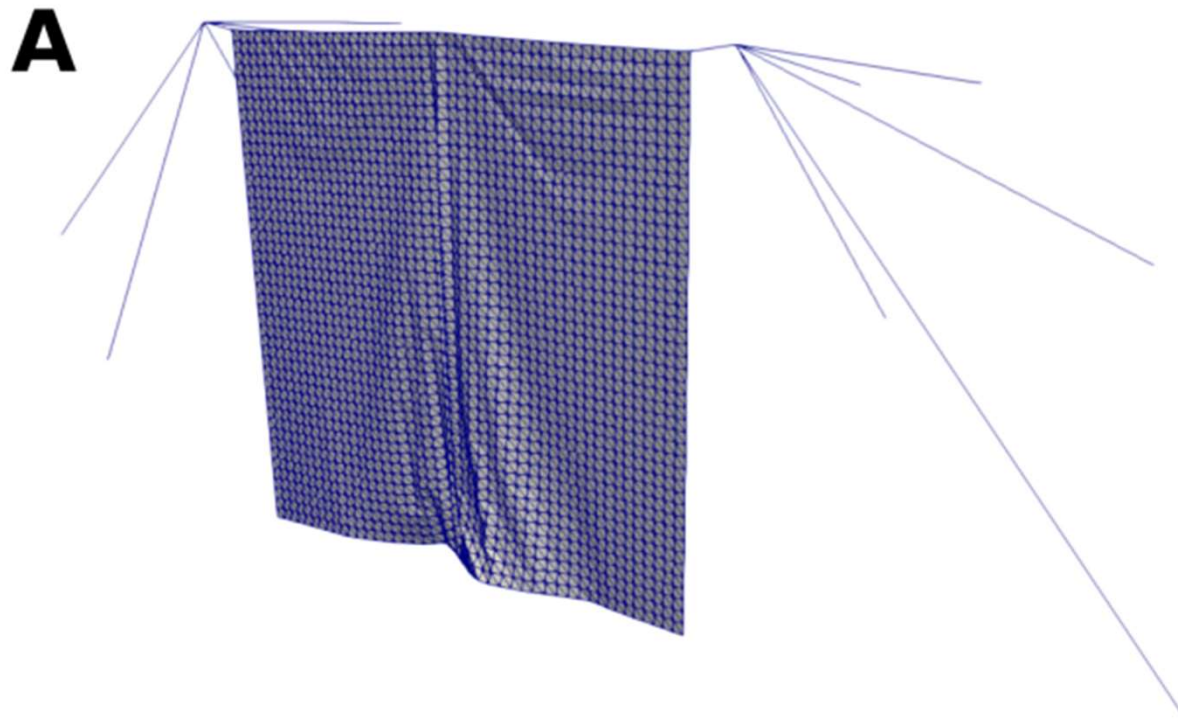
$$\tilde{\mathbf{C}} = \frac{1}{1 - \nu_{xy}\nu_{yx}} \begin{bmatrix} E_x & \nu_{xy}E_x & 0 \\ \nu_{yx}E_y & E_y & 0 \\ 0 & 0 & (1 - \nu_{xy}\nu_{yx})G \end{bmatrix},$$

$$\frac{\nu_{xy}}{E_y} = \frac{\nu_{yx}}{E_x}.$$

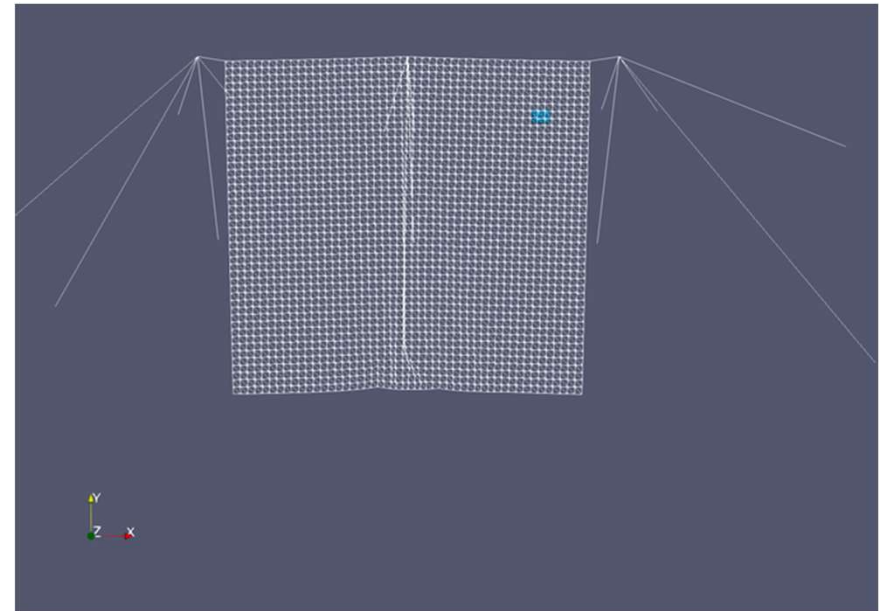
Anisotropic material law
Münsch & Reinhardt



Attenuator



Rock fall –Protection



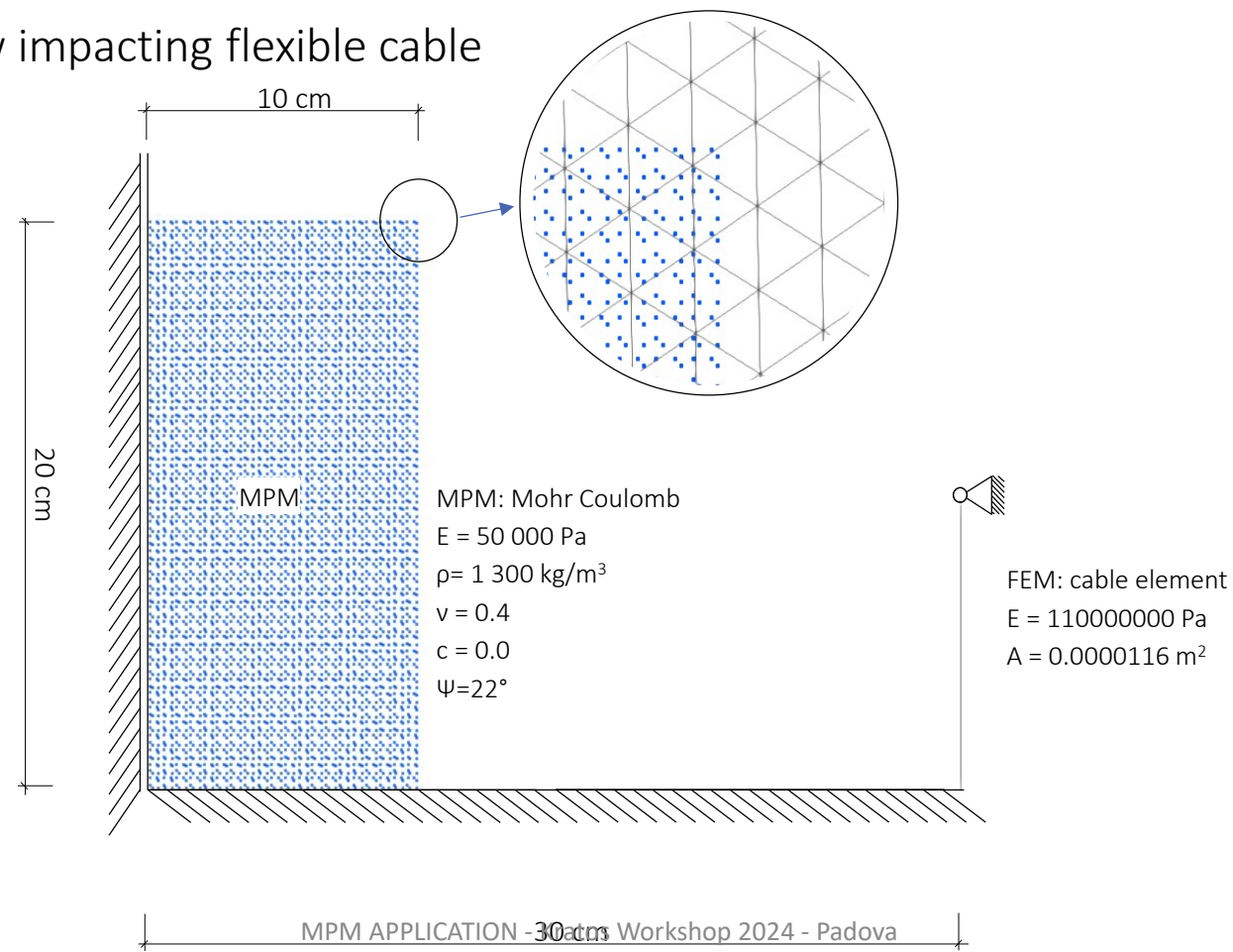
Curtesy of Klaus
Sautter, TUM and
Geobrugg



OUR CHALLENGE:

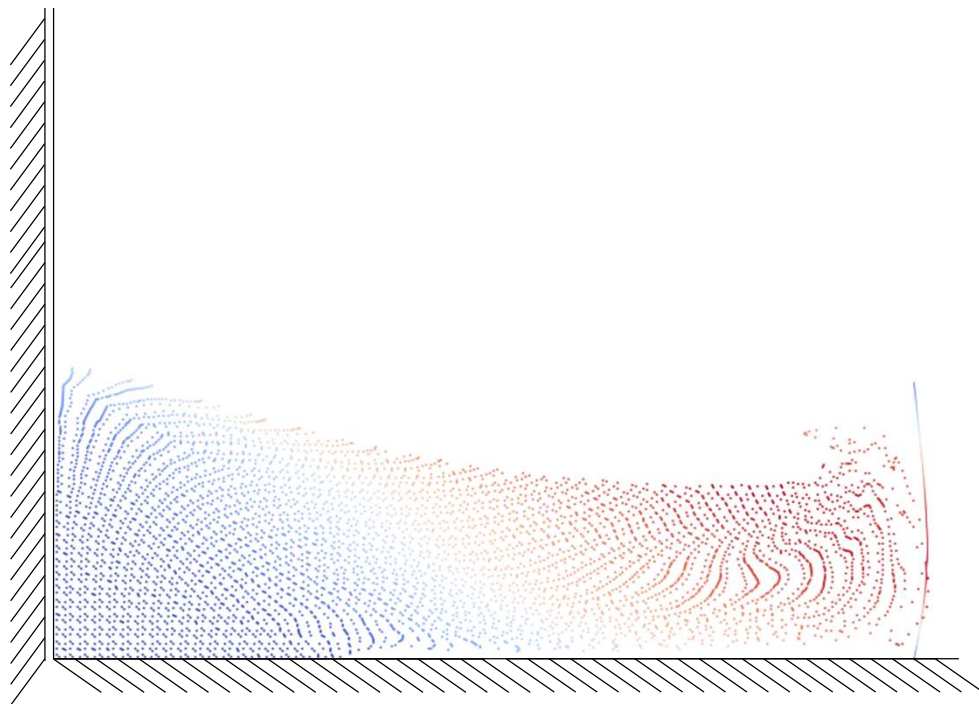


Example: Granular flow impacting flexible cable

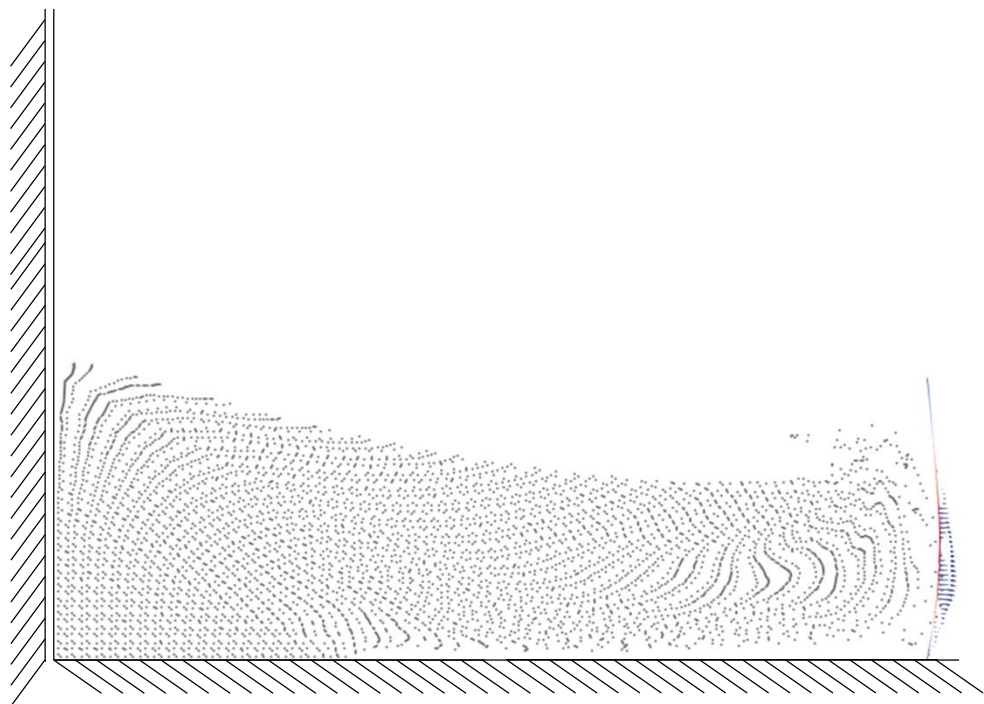


Example: Granular Flow impacting flexible Cable

Results:



Contact forces



3D example: Granular flow impacting flexible membrane

MPM: Mohr Coulomb

$E = 50\,000\text{ Pa}$

$\rho = 1\,300\text{ kg/m}^3$

$\nu = 0.4$

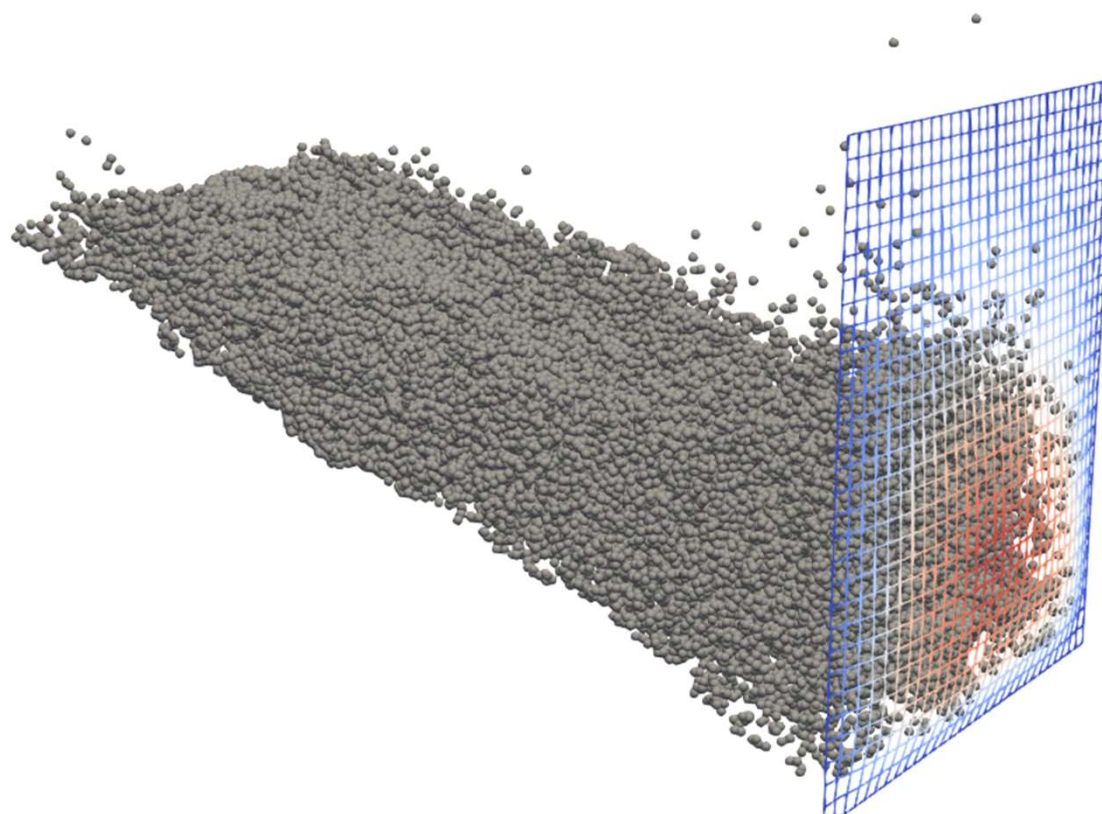
$c = 0.0$

$\Psi = 22^\circ$

FEM: Membrane

$E = 100\text{ MPa}$

$t = 0.86\text{ cm}$



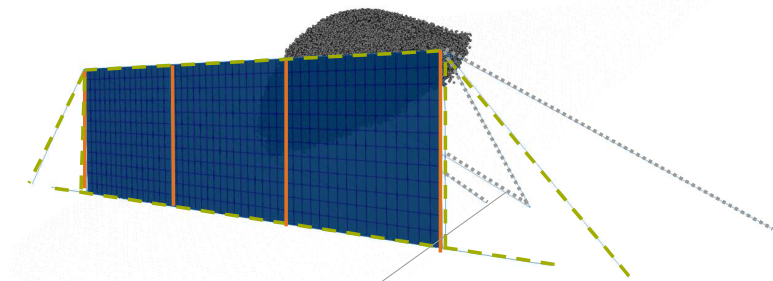
Protective structure



a) Debris-flow event in Japan

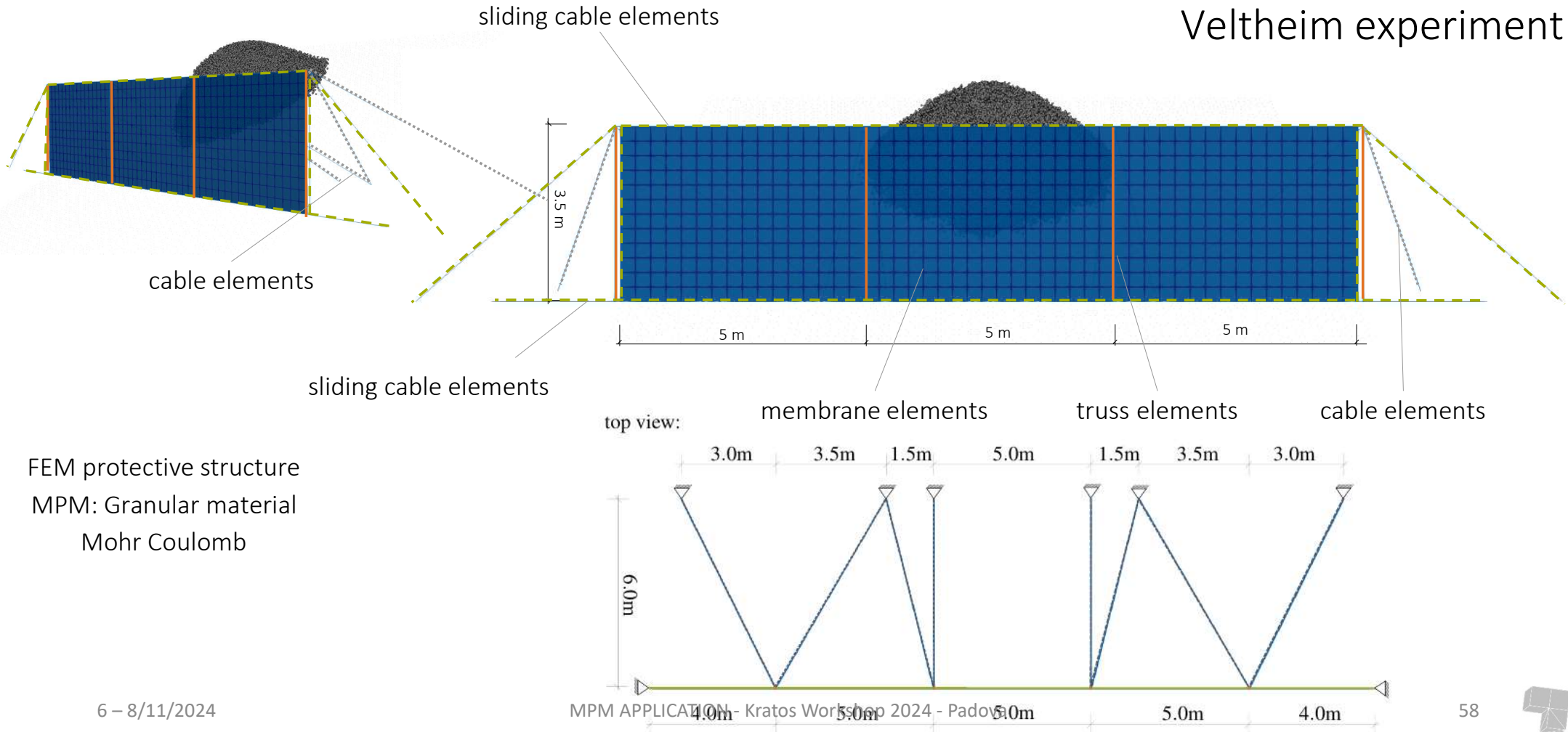


b) Mud-flow experiments in Switzerland

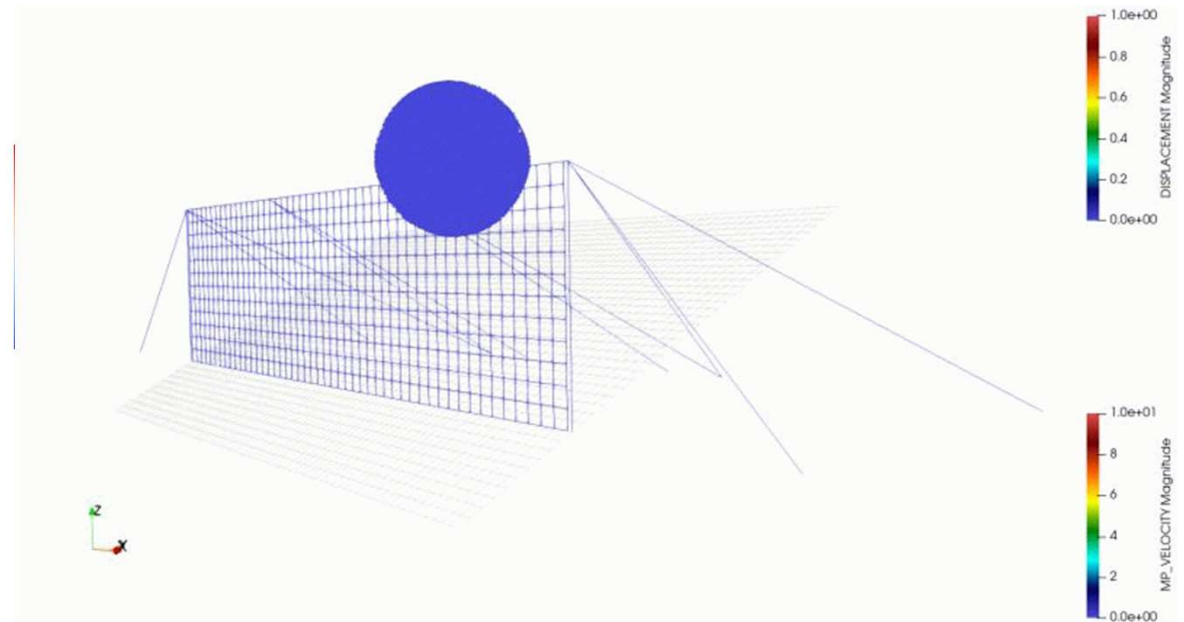
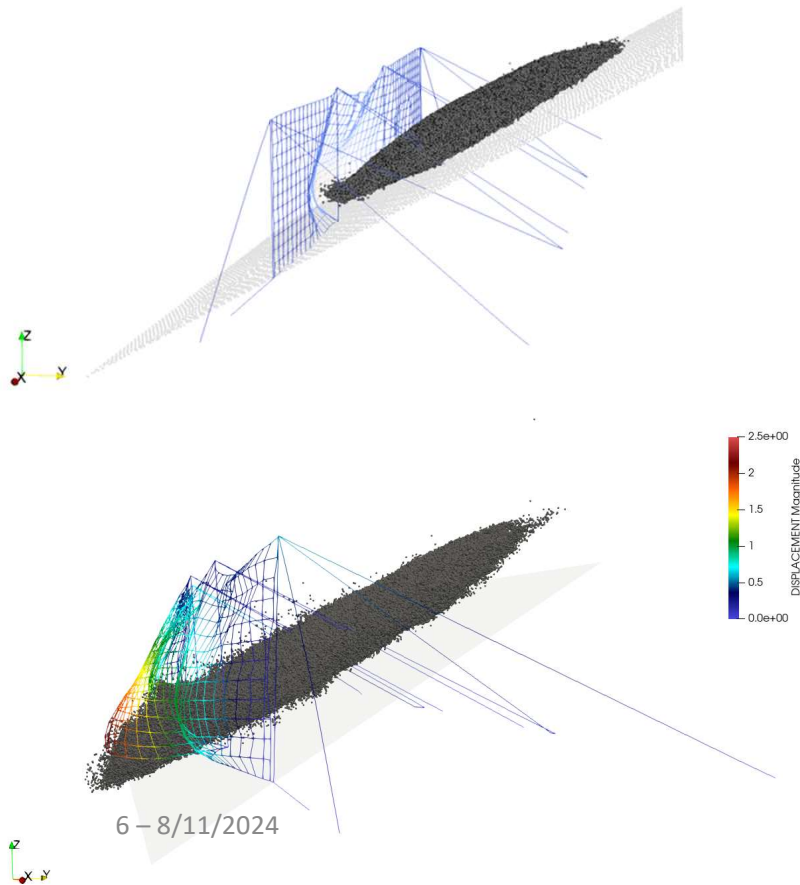


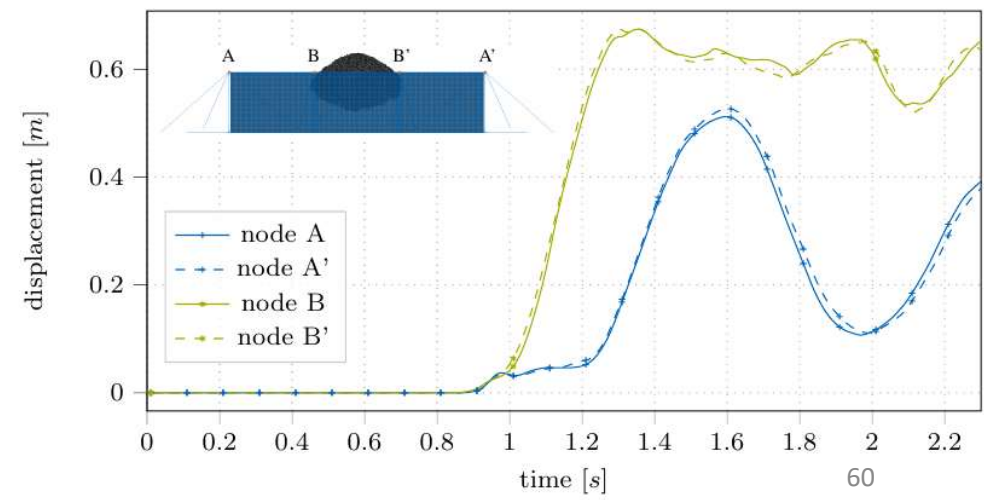
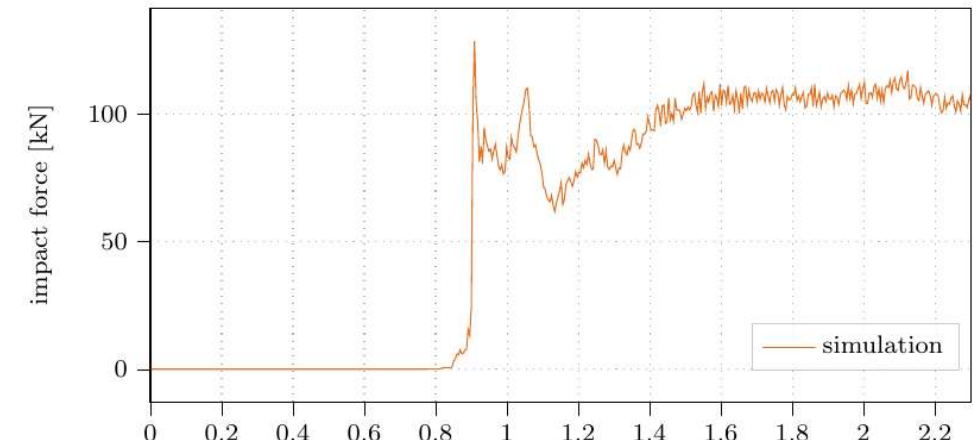
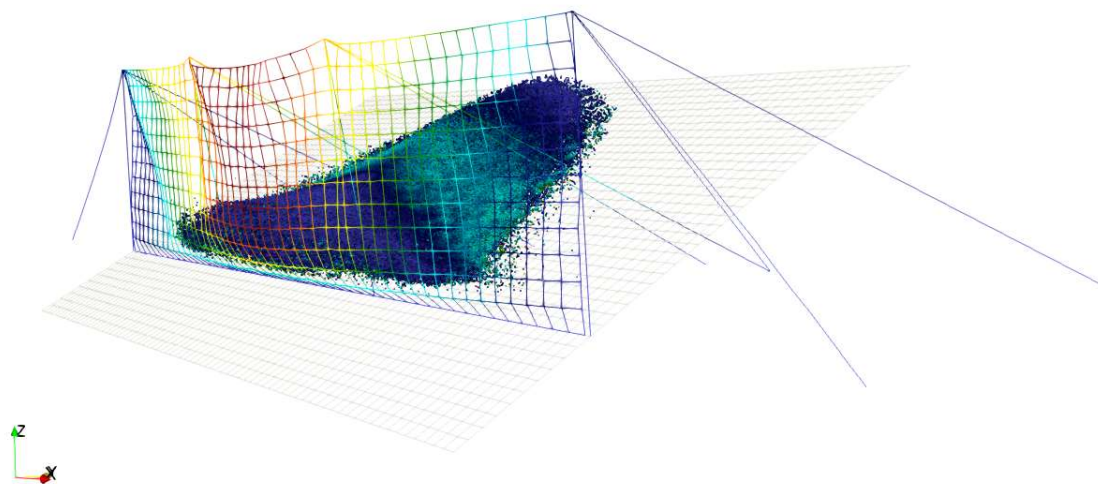
PARTITIONED MPM – FEM COUPLING

Veltheim experiment



Mud flow protection

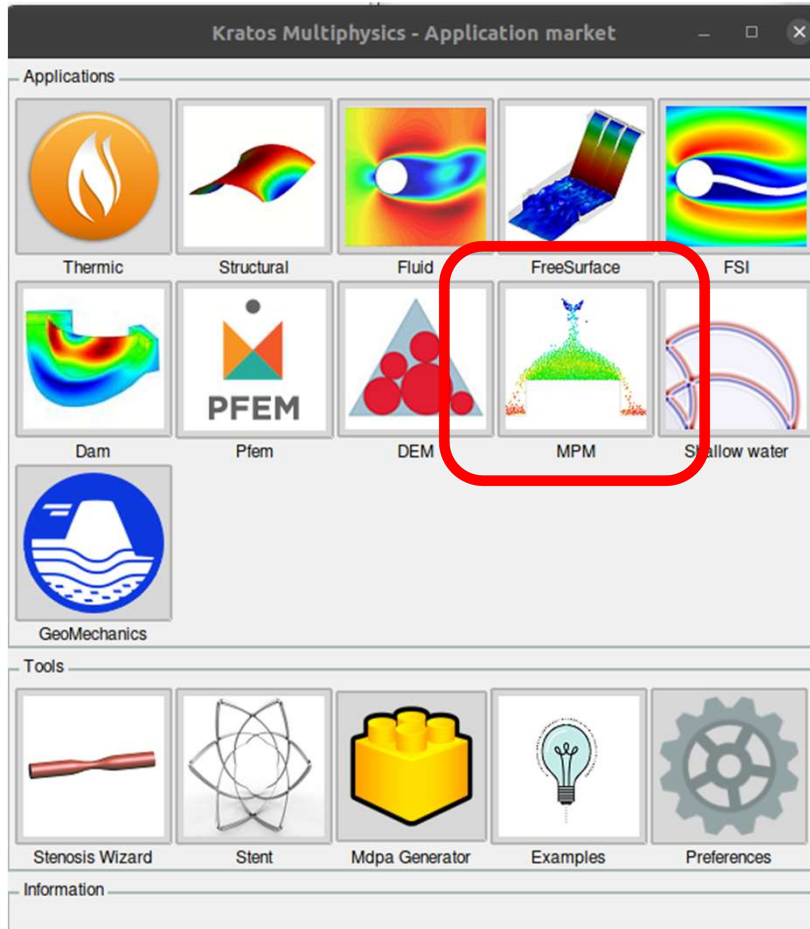




MPM application



Tutorials and examples



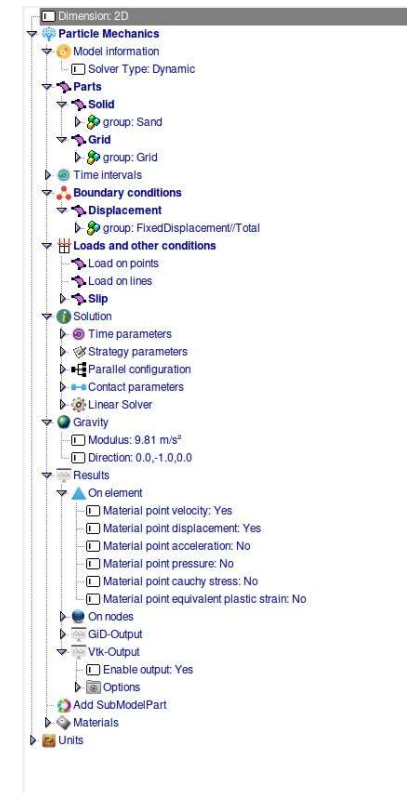
GUI for MPM

PRE-PROCESS

- GiD problemtype

POST-PROCESS

- GiD post
- Paraview



Tutorials and examples

The screenshot displays the 'MPM Application' page on the 'Kratos Multiphysics Documentation' website. The page features a blue header with navigation links: Home, Nav, GitHub, News, Kratos, Applications, and Feedback. A sidebar on the left contains a menu for 'MPM Application' with options: General (selected), MPM Application, MPM example in GiD, and Examples. The main content area has a title 'MPM Application' and a 'Table of Contents' with links to Getting Started, Examples, Theory, Features, License, and Contact. Below this is an 'Edit me' button and a paragraph describing the application: 'This application implements the Material Point Method (MPM) with main motivations of simulating non-linear materials, such as free-surface flows, geomechanical phenomena, and extreme events involving impact, fragmentation, blast, multi-phase interaction, failure evolution, etc.' A QR code is located on the left side of the page. At the bottom, there is a visualization of a simulation showing a grid of material points and a blue vertical bar on the right.

Kratos Multiphysics Documentation

Nav GitHub News Kratos Applications Feedback

MPM Application

- General
- MPM Application
- MPM example in GiD
- Examples

MPM Application

Table of Contents

- Getting Started
- Examples
- Theory
- Features
- License
- Contact

Edit me

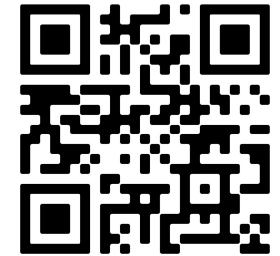
This application implements the Material Point Method (MPM) with main motivations of simulating non-linear materials, such as free-surface flows, geomechanical phenomena, and extreme events involving impact, fragmentation, blast, multi-phase interaction, failure evolution, etc.

MPM Application

- General
- MPM example in GiD
- Introduction
- Set up of example
- Meshing
- Calculation
- Post processing
- Examples



MPM application - features



Formulations:

- Irreducible formulations (U displacement based)
- Mixed UP formulations

Element types:

- Updated Lagrangian elements - triangular and quadrilatera (2D) and tetrahedral and hexahedral (3D), structured and unstructured, using classical or partitioned quadrature rules (this latter limited to explicit MPM)
- Updated Lagrangian axis-symmetric elements - triangular and quadrilateral (2D), structured and unstructured
- Updated Lagrangian mixed UP elements - triangular (2D) and tetrahedral (3D), structured and unstructured, stabilized using Variational Multiscale Stabilization or Pressure Projection techniques



MPM application - features

Constitutive laws:

- Linear isotropic elastic materials - plane strain, plane stress, axis-symmetric, and 3D
- Hyperelastic Neo-Hookean laws - finite strain, plane strain, axis-symmetric, and 3D
- Displacement based Newtonian fluid

Elasto-plastic laws:

- Mohr Coulomb - finite strain, associative and non-associative, plane strain, axis-symmetric, and 3D
- Mohr Coulomb with Strain Softening - finite strain, associative and non-associative, plane strain, axis-symmetric, and 3D

Critical state laws:

- Modified Cam-Clay - finite strain, plane strain, axis-symmetric, and 3D
- Johnson Cook Thermal Plastic (just for explicit MPM)

Natural interface with other constitutive laws present in Kratos applications (geomechanics/ structural)



MPM application - features

BOUNDARY CONDITIONS

Grid-Based Conditions (conforming): applied directly on the background nodes

- Neumann: Point load
- Neumann: Line load (a distributed load applied over a line)
- Neumann: Surface load (a distributed load applied over a face)
- Dirichlet: Slip and No slip condition on arbitrary boundary.

Particle-Based Conditions (non conforming): applied in moveable boundary particles

- Neumann: Moving point load
- Dirichlet: Imposition of displacements (homogeneous and inhomogeneous) using penalty method



MPM application - features

Strategies and schemes:

- Implicit - Newmark/Bossak prediction and correction scheme for static, quasi-static, and dynamic problems
- Explicit

Other features:

- Partitioned coupling with Finite Element Method - weak and strong coupling of nonconforming discretization
- Partitioned coupling with the Discrete Element Method
- Particle erase features - to delete particle outside the interest domain



The MPM team in KRATOS



ANTONIA



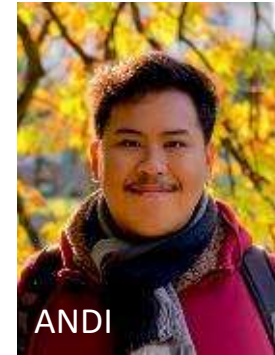
VERONIKA



LAURA



NICOLÒ



ANDI



ELEONORA

- Non conforming BCs
- MPM-DEM
- MPM-FEM
- Mixed stabilized MPM
- CFD in MPM
- MPM architecture
- MPI with MPM
- MPM - SW
- Surrogate MPM models





 <https://github.com/KratosMultiphysics>

 www.youtube.com/@kratosmultiphysics3578

